Thricing Is All You Need: A Sixth-Order Kernel Estimator of the Risk-Neutral Density from a Single Option Chain

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Abstract

The risk-neutral density of an index at a fixed expiry is the second strike derivative of the call price curve, but a listed chain supplies that curve only at a finite, noisy set of strikes. This paper proposes an estimator that works on the normalized call, which is the survival function of the integrated-tail distribution of the underlying. That function is fitted by a cubic smoothing spline and smoothed by a Gaussian kernel; a three-bandwidth generalized jackknife, which we call thricing, raises the order of the kernel to six and makes the Gaussian convolution of the spline, and hence the density and the fitted call, available in closed form, and the same device applied to the smoothing spline keeps that order when the penalty is on. One number read from the chain, the roughness of the thriced interpolant, sets both tuning constants in closed form: it widens a bandwidth at the $n^{-1/9}$ rate of a second-derivative estimator, and it sets the spline penalty. We prove in-fill consistency under sub-Gaussian noise in the quotes, with the data-dependent bandwidth and penalty. In a Monte Carlo study with Heston, variance-gamma, and Bates targets, three noise levels, and three strike meshes, with every competing method tuned by one common criterion, the estimator at its rule has the smallest integrated squared error, or one statistically indistinguishable from the smallest, in every Heston and Bates design at the central noise level and in the listed-shaped variance-gamma design, and it comes within a factor of two of its own oracle bandwidth. At infeasible oracle bandwidths the implied-volatility smoothers are more accurate on regular meshes, so the contribution is a complete procedure whose tuning works

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rather than a kernel that dominates. On listed S&P 500, Nasdaq-100, and Russell 2000 chains it has the smallest out-of-sample pricing error of the competing methods on every chain, measured in index points, in units of the quoted half-spread, and in implied volatility, while delivering a unimodal density with unit mass. A sensitivity analysis shows that the conclusions do not depend on the bandwidth constant.

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1. Introduction

Breeden and Litzenberger (1978) showed that the risk-neutral density of the underlying at a fixed expiry equals the second strike derivative of the European call price curve, scaled by the discount factor. The identity requires a twice differentiable price curve, whereas a market supplies a finite ordered list of strikes and a noisy quote at each. Numerical second differentiation of such a list is ill-posed: interpolants oscillate, their second derivatives change sign, and the result depends on the interpolation scheme. The literature responds with parametric mixtures and Gram–Charlier expansions, implied-volatility interpolants, local polynomials of the call, and the positive convolution approximation of Bondarenko (2003); Section 2 reviews these approaches.

This paper proposes a nonparametric estimator that uses the quotes of one maturity and nothing else. Its construction rests on an elementary observation. The call normalized by the dividend-adjusted spot, $\tilde{C}(K) = C(K)/(S_0 e^{-qT})$, decreases from 1 at $K = 0$ to 0 at $K = \infty$: it is the survival function of a probability distribution on $(0, \infty)$, namely the integrated-tail (equilibrium) distribution of S_T under the risk-neutral measure, whose density is the discounted tail probability $(1 - F_{\mathbb{Q}}(K))/F$. Its characteristic function is $(\varphi_{\mathbb{Q}}(s) - 1)/(\mathrm{i}sF)$, so a model that supplies $\varphi_{\mathbb{Q}}$ prices calls by one Fourier inversion (Lewis, 2000), and conversely the quotes determine the characteristic function of S_T. The estimator inverts this identity: it fits the integrated-tail distribution function $P = 1 - \tilde{C}$ by a cubic smoothing spline, completes the unquoted wings, and smooths the derivative of the spline by a Gaussian kernel. Differentiating the smoothed curve once more gives the density, and integrating it once gives the fitted call, so the density and the call are two readings of the same object. Both are available in closed form

because every piece of the spline is a polynomial.

A Gaussian kernel is of order two and leaves a bias of order h^2. Twicing (Tukey, 1977; Schucany and Sommers, 1977) removes that term by combining two bandwidths; we combine three, which removes the h^4 term as well and gives an order-six kernel. We call this construction thricing. It is a generalized jackknife in the sense of Schucany and Sommers (1977) and Jones and Foster (1993), and we do not claim it is new in principle; its merit here is that it leaves the estimator in closed form, since the thriced kernel is a signed mixture of three Gaussians. Because P is two derivatives smoother than the density, the leading bias of a Gaussian smooth of P is large relative to its variance at the bandwidths that a second derivative calls for, and the higher-order correction pays off in practice: on exact Heston quotes it reduces the integrated squared error by three orders of magnitude on a dense chain.

Two tuning choices remain, and both are rule based, driven by one number read from the chain: the roughness of the thriced interpolant, the excess of its positive-part total variation over that of the thriced kernel itself. It is zero on exact quotes and grows with the noise. The bandwidth is a bounded multiple of $n^{-1/9}$, the rate that balances bias and variance of a second-derivative estimator, scaled by the forward times the at-the-money implied volatility times the root of maturity, and widened by a power of one plus the roughness. The spline penalty is zero when the roughness is zero and otherwise the closed-form value at which the spline attenuates the frequency $1/h$ by one over one plus the roughness. Because the smoothing spline does most of the smoothing once the penalty is on, and a cubic smoothing spline is a fourth-order smoother, the spline is thriced as well: it is fitted to the knots and twice more to its own residuals, which raises its order and leaves the sixth-order kernel in charge of the bias. Section 3.5 states both rules and explains every constant; Section 5.4 shows that the results are insensitive to the bandwidth constant across the range 0.27 to 0.60, and that the penalty matters only on noisy chains. Theorem 1 gives in-fill consistency under sub-Gaussian noise in the quotes, with the data-dependent bandwidth and penalty, whenever the strike mesh is quasi-uniform, $h \to 0$ and $nh^5/\log n \to \infty$.

The empirical contribution is a comparison under a protocol in which no method is handicapped by its tuning. Every estimator that has a bandwidth—the positive convolution approximation of Bondarenko, the implied-volatility kernel of Aït-Sahalia and Lo (1998), the shape-constrained local linear of Aït-Sahalia and Duarte (2003), the local cubics of Grith et al. (2012), and our own—is tuned by the same criterion, even/odd out-of-sample pricing error on one geometric bandwidth grid, and the tuning is nested inside the hold-out. The rules that the source pa-

pers propose are reported next to the tuned versions. Pricing errors are measured in index points, in units of the quoted half-spread, and in implied volatility, so that chains of different levels can be averaged and so that an error can be judged against the market's own uncertainty. Three targets (Heston, variance gamma, and the Bates jump diffusion), three noise levels, and three strike meshes enter the Monte Carlo study, with 200 replications and standard errors. On the listed side, the four chains of the main table are complemented by every remaining standard monthly expiry in the same snapshots, 31 chains in all, scored with the rules held fixed.

The results are as follows. On exact quotes, tails and thricing reduce the error of the plain spline by several orders of magnitude, and the rule bandwidth is close to the oracle on sparse chains. With noise, the estimator at its rule has the smallest mean integrated squared error, or one within two standard errors of the smallest, in seven of nine designs at the central noise level, and at the other two noise levels in every Heston and Bates design but one; the exceptions are the peaked variance-gamma density on the two regular meshes, where the positive convolution approximation and the implied-volatility kernel of Aït-Sahalia and Lo are better by factors of 1.4 to 2.3. An oracle experiment separates the estimator from its tuning: at the bandwidth that minimizes the error of each draw, which is infeasible, the implied-volatility smoothers are two to six times more accurate than ours on the regular meshes, while on the listed-shaped mesh the methods are close. The rule bandwidth brings our estimator within a factor of two of its oracle on every design; no feasible rule in the comparison, cross-validated or published, brings the implied-volatility smoothers within a factor of forty of theirs on the dense mesh. What the paper offers is therefore a procedure that works as published, not a kernel that dominates at every bandwidth. On the listed chains the estimator has the smallest nested hold-out error of the competing methods on each of the four main chains under all three metrics, and is within four thousandths of a half-spread of its own cross-validated version, while keeping one mode, unit mass, and no negative mass to three decimals; its advantage over the next method is between a factor of about 1.3 and four in half-spread units. Every good method prices inside the quoted spread, so the economic content of the listed comparison is in the density diagnostics as much as in the pricing numbers. We also find that pricing cross-validation under-smooths on dense noisy chains for every method, including ours, which is an argument for a rate-based rule.

The paper is organized as follows. Section 2 reviews related work. Section 3 constructs the estimator and states the tuning rules. Section 4 gives the asymptotic result. Section 5 reports the simulation study and the sensitivity analysis.

Section 6 describes the data, the comparison protocol, and the listed results. Section 7 concludes. The proof, the computational details, and the full tables are in the appendices.

2. Related Work

The identity of Breeden and Litzenberger (1978) is the continuous-strike version of a spanning argument of Ross (1976): with a put at every strike, the second derivative of the put curve is a density and any sufficiently regular payoff can be synthesized. Dupire (1994) obtained the analogous result in calendar time. Observed markets supply neither a continuum of strikes nor noiseless quotes.

A first family of methods specifies the density of S_T parametrically and fits the implied prices to the quotes. Ritchey (1990) and Melick and Thomas (1997) use mixtures of lognormals, Bahra (1997) and Söderlind and Svensson (1997) related specifications for interest-rate and foreign-exchange options. Two or three components suffice for a unimodal skewed density, but the fit is poorly identified once more components are admitted, and the approximation to a stochastic-volatility or jump-diffusion density need not improve as more quotes arrive. Expansion methods perturb a baseline density by Gram–Charlier or Hermite terms (Jarrow and Rudd, 1982; Corrado and Su, 1996; Rompolis and Tzavalis, 2007; Xiu, 2014); they match a finite number of moments but are not guaranteed to stay nonnegative.

A second family, the most common in applied work, interpolates the Black–Scholes implied-volatility function and differentiates the implied call twice. Shimko (1993) fitted a quadratic in strike, Malz (1997) a cubic spline in delta, Bliss and Panigirtzoglou (2002) a smoothing spline, and Figlewski (2010) completes the tails with a generalized extreme-value distribution. Implied volatility is a flat function of strike, so a low-order interpolant has small bias when quotes are accurate; the drawback is that the transformation inflates price noise where vega is small, and an interpolating spline then treats noisy tail quotes as exact. Jackwerth (2004) and Figlewski (2018) note that a cubic spline of implied volatility puts kinks in the second derivative and recommend higher-degree splines. Shape-constrained smoothers of the surface (Fengler, 2005, 2009; Gatheral and Jacquier, 2014; Ludwig, 2015) and neural smoothers (Ackerer et al., 2020) address the sign problem at the cost of a more involved optimization.

A third family estimates the call, or its implied volatility, by kernel regression or local polynomials and differentiates. Aït-Sahalia and Lo (1998) use a Nadaraya–Watson kernel on 14,431 S&P 500 options pooled across 1993, because a single expiry was treated as too short a list; Aït-Sahalia and Lo (2000)

recovered an implied risk-aversion function from the same estimator. Aït-Sahalia and Duarte (2003) work one day at a time on twenty to fifty strikes and impose monotonicity and convexity of the call; they report that unconstrained local polynomial estimates of the density took negative values on nearly every day of 1999 and that the raw quotes violated convexity on 24% of the prints. Grith et al. (2012) survey this literature and estimate a local cubic of the rescaled call with a quartic kernel; they find the cross-validated bandwidth too wiggly and plot their densities at twice that width. Song and Xiu (2016) use a local linear smoother of the implied volatility of a single maturity on a large panel. Kernel estimates of regression derivatives go back to Gasser and Müller (1984); the optimal bandwidth for a second derivative is of order $n^{-1/9}$ (Fan and Gijbels, 1996). Fan and Mancini (2009) combine a parametric guide with a nonparametric correction, and Yatchew and Härdle (2006), Birke and Pilz (2009) and Monteiro et al. (2008) study shape-constrained least squares.

Bondarenko (2003) proposed the positive convolution approximation (PCA): the density is the convolution of a positive kernel with a discrete probability chosen to minimize the squared pricing error. The result is a density by construction, and in his comparison it dominated mixtures, expansions and several regularization methods; Lu (2019) report the same ranking on affine jump-diffusions. Rubinstein (1994), Jackwerth and Rubinstein (1996), Stutzer (1996) and Buchen and Kelly (1996) recover probabilities on a grid by minimizing a distance to a prior under pricing constraints; Cont and Tankov (2004) and Monnier (2014) treat the problem as a linear inverse problem.

The estimator of this paper is a kernel smooth of the derivative of a spline fitted to the integrated-tail distribution function of one maturity, with a sixth-order Gaussian jackknife kernel. Comparisons of methods on market data use pricing error or stability as the criterion (Cooper, 1999; Jondeau and Rockinger, 2000; Coutant et al., 2001; Anagnou et al., 2005; Bliss and Panigirtzoglou, 2002; Liu et al., 2007); comparisons on a known process are rarer (Bondarenko, 2003; Lai, 2014; Santos and Guerra, 2015; Lu, 2019). Section 5 is of the second kind and Section 6 of the first. Relative to those studies, the novelty of our comparison is the tuning protocol: every method is tuned by the same out-of-sample criterion, the source papers' own rules are shown alongside, and the errors are expressed in half-spreads and implied volatility.

3. The Estimator

3.1. The call as the survival function of the integrated-tail distribution

Assume a risk-neutral measure $\mathbb{Q}$ under which S_T has a density $f_{\mathbb{Q}}$ on $(0,\infty)$ with $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$. Let r be the continuously compounded rate, q the dividend yield, and $F = S_0 e^{(r-q)T} = \mathbb{E}^{\mathbb{Q}}[S_T]$ the forward. The call curve is $C(K) = e^{-rT}\int_K^\infty (x-K) f_{\mathbb{Q}}(x)\,\mathrm{d}x$, and we define the normalized call

$$\tilde{C}(K) \;\equiv\; \frac{C(K)}{S_0 e^{-qT}} \;=\; \frac{1}{F}\int_K^\infty (x-K)\, f_{\mathbb{Q}}(x)\,\mathrm{d}x.$$

$\tilde{C}$ is convex, decreasing, equal to 1 at $K=0$ and to 0 at infinity, so $P \equiv 1-\tilde{C}$ is a distribution function on $[0,\infty)$ with density

$$P'(K) \;=\; \frac{1-F_{\mathbb{Q}}(K)}{F}, \qquad P''(K) \;=\; -\frac{f_{\mathbb{Q}}(K)}{F}. \tag{1}$$

P is the integrated-tail, or equilibrium, distribution of S_T. Writing $\varphi_{\mathbb{Q}}(s) = \mathbb{E}^{\mathbb{Q}}[e^{\mathrm{i}sS_T}]$ and $\varphi_P(s) = \int e^{\mathrm{i}sK}\,\mathrm{d}P(K)$, an integration by parts gives, for $s \neq 0$,

$$\varphi_P(s) \;=\; \frac{\varphi_{\mathbb{Q}}(s)-1}{\mathrm{i}s\,F}, \qquad \varphi_{\mathbb{Q}}(s) \;=\; 1 + \mathrm{i}s\,F\,\varphi_P(s). \tag{2}$$

Read from left to right, (2) is a pricing formula (Lewis, 2000). This paper reads it from right to left: φ_P, and hence P, is estimated from the quotes, and the density follows from (1) as $f_{\mathbb{Q}} = -FP''$. The Fourier form is useful because smoothing by a Gaussian kernel and differentiation are both multipliers of φ_P, and because the pricing direction supplies the exact quotes of Section 5.

3.2. Smoothing spline and Gaussian smoothing

Observed calls at strikes $0 < K_1 < \cdots < K_m$ supply $P_i = 1 - C_i/(S_0 e^{-qT})$. The estimator fits the natural cubic smoothing spline P^{s} that minimizes

$$\sum_i \big(P_i - g(K_i)\big)^2 \;+\; c\,F^3 \int \big(g''(K)\big)^2\,\mathrm{d}K, \tag{3}$$

over the knots described below. The factor F^3 makes c dimensionless. At $c=0$ the spline interpolates. Let κ_h be the Gaussian density with standard deviation h and $\mathcal{K}_h$ its distribution function. The density and the fitted distribution function are

$$\hat{f}_{\mathbb{Q}}(K) \;=\; -F\,\big(P^{\mathrm{s}\prime} * \kappa_h\big)'(K), \qquad \hat{P}(K) \;=\; \big(P^{\mathrm{s}\prime} * \mathcal{K}_h\big)(K), \tag{4}$$

and the fitted call is $\hat{C} = S_0 e^{-qT}(1-\hat{P})$. In the Fourier domain (4) multiplies φ_{P^s} by $e^{-h^2 s^2/2}$ and by is. In the strike domain, each knot interval of $P^{s\prime}$ is a quadratic, so each term of (4) is an integral of a quadratic against a Gaussian and is elementary; Appendix B describes three ways of evaluating the sum, the fastest of which takes a few milliseconds on a chain of four hundred strikes.

At $c = 0$, (4) is a kernel estimator of the second derivative of P in the sense of Gasser and Müller (1984), with the interpolating spline in place of the step function. The spline matters: a piecewise-linear interpolant has a derivative with jumps at the knots, whose convolution with κ_h' oscillates at the scale of the mesh unless h is many knot spacings wide. The cubic spline has a continuous second derivative, so the convolution is smooth on the scale of h whenever h exceeds about one knot spacing, and the bias is governed by the kernel rather than by the mesh.

3.3. Completing the tails

The spline stops at the first and last quotes, and P is a distribution function on $(0, \infty)$. On $(0, K_1]$ a deep in-the-money call is intrinsic, $\tilde{C}(K) = 1 - K/F$, so $P(K) = K/F$ is a straight line through the origin. The estimator includes the knot $P(0) = 0$ and otherwise lets the spline pass through $P(K_1)$; when the first quote is at intrinsic the fitted segment is exactly K/F. On the right, if the last quoted call C_m is positive the residual call is continued linearly, with the slope of the last strictly negative step of the quoted call clipped to $[-e^{-rT}, 0)$, until it reaches zero at $K_{\text{end}} \le 8F$, and the knot $P(K_{\text{end}}) = 1$ is added. A flat last step, two quotes on the premium floor, does not count as a decay. If $C_m = 0$ the right wing is empty. The tail knots enter (3) like the quoted ones.

The tails matter for the second-derivative estimate even far from the wings. A natural spline imposes $P'' = 0$ at its end knots, which lie inside the strike window when only the quotes are used; the completion moves those boundary conditions to 0 and to K_{end}. The higher-order kernel of the next section has negative sidelobes of width proportional to h, and on a truncated strip those sidelobes amplify the missing mass. Table 1 quantifies the effect: on a sparse chain the error of the quoted spline falls by four orders of magnitude when the wings are added. The linear continuation is a convention, and a crude one in the limit: on exact quotes of the listed-shaped mesh of Section 5.1, whose last strike lies inside the evaluation window, a lognormal wing at the implied volatility of the last quotes lowers the error by a factor of three to a thousand. We nevertheless keep the line, because under noise the lognormal wing inherits the error of the wing quotes it is fitted to,

whether one or ten of them, and on every noisy design and listed chain we tried it raised the error.

3.4. Thricing: a sixth-order Gaussian jackknife kernel

The Gaussian kernel has second moment h^2, so the bias of (4) is of order h^2. The generalized jackknife of Schucany and Sommers (1977) removes the leading term by combining estimates at two bandwidths; Tukey (1977) called the simplest such combination twicing. Replacing $e^{-h^2 s^2/2}$ by $2e^{-h^2 s^2/2} - e^{-h^2 s^2}$, the transform of $2\kappa_h - \kappa_{h\sqrt{2}}$, cancels the second moment and leaves a bias of order h^4. Taking the construction one step further, the combination

$$\tfrac{8}{3}\, e^{-h^2 s^2/2} - 2\, e^{-h^2 s^2} + \tfrac{1}{3}\, e^{-2h^2 s^2} \tag{5}$$

is the transform of $L_h = \frac{8}{3}\kappa_h - 2\kappa_{h\sqrt{2}} + \frac{1}{3}\kappa_{2h}$, whose moments of orders two and four vanish. We call this thricing. L_h is a sixth-order kernel in the sense of Jones and Foster (1993), and the estimator becomes

$$\hat{f}^{\mathrm{th}}_{\mathbb{Q}} \;=\; \tfrac{8}{3}\, \hat{f}_{\mathbb{Q},h} - 2\, \hat{f}_{\mathbb{Q},h\sqrt{2}} + \tfrac{1}{3}\, \hat{f}_{\mathbb{Q},2h}, \tag{6}$$

with the same combination applied to $\hat{P}$. The bias is of order h^6 when $f_{\mathbb{Q}}$ has a bounded sixth derivative. Because L_h takes negative values, $\hat{f}^{\mathrm{th}}_{\mathbb{Q}}$ is not a density by construction; Sections 5 and 6 report its negative mass, which is zero to three decimals on every listed chain. The fitted $\hat{P}^{\mathrm{th}}$ is clipped to $[0, 1]$ before pricing and premiums are floored at zero.

The correction is large in this problem for a specific reason. The call is a double integral of the density, so P is two derivatives smoother than $f_{\mathbb{Q}}$, and the bias of a kernel smooth of P at the bandwidth that a second derivative requires, $h \asymp n^{-1/9}$, dominates the variance of order $(nh^5)^{-1}$. On exact quotes that bias is the entire error; twicing removes its leading term and thricing the next. Jackknifing is not free: $\int |L_h| = 1.196$, so the variance constant rises by about 43% relative to the Gaussian, and the sixth-order bias constant involves the sixth derivative of $f_{\mathbb{Q}}$. Table 1 and Table 3 show that the trade is favourable at the bandwidths the rule selects.

Thricing the spline. On noisy quotes the penalty of Section 3.5 is positive, and the smoothing spline then smooths as well as the kernel. On a uniform mesh of spacing δ the cubic smoothing spline multiplies a Fourier mode of frequency ω by $1/(1 + x)$ with $x = \lambda\delta\omega^4$ (Silverman, 1984; Green and Silverman, 1994), an attenuation whose expansion $1 - x + \cdots$ makes the spline a fourth-order smoother:

its bias in P'' is of order β^4, with $\beta = (\lambda\delta)^{1/4}$ the half-width of its equivalent kernel. Whenever the penalty is on, β exceeds h (Table 6), so that fourth-order bias, not the sixth-order one of L_h, would govern the estimate, and the penalty $\int (P'')^2 = \int f_{\mathbb{Q}}^2 / F^2$ shrinks the density toward zero, which flattens its peak. The remedy is the device of Tukey (1977) once more, now applied to the spline: fit the smoothing spline, fit it again to the residuals, and again to the residuals of that fit, and add the three. In operator form the thriced spline is

$$P_3^{\mathrm{s}} = (I - (I - S_\lambda)^3)P, \qquad \text{attenuation} \quad 1 - \Big(\frac{x}{1+x}\Big)^3 = 1 - x^3 + \cdots, \tag{7}$$

where S_λ is the smoothing-spline operator. Each pass is a cubic spline on the same knots, so their sum is one, and the closed-form convolution of Section 3.2 applies unchanged. The attenuation is now of order twelve, so the sixth-order bias of the kernel is again the leading one; the price is the usual one, a variance constant that rises from 0.088 to 0.24 at equal β for the second derivative, which the penalty rule below offsets by a larger β. Thricing the spline leaves exact quotes untouched, because the penalty is zero there and the three passes reduce to the interpolant.

3.5. The bandwidth and penalty rules

Two constants have to be chosen: the bandwidth h in (6) and the penalty c in (3). Both are set by rules, not by cross-validation, because Section 6 shows that pricing cross-validation under-smooths a density estimator, and because a rule makes the estimator deterministic given the chain. Both rules read one number from the chain, the roughness of the thriced interpolant, which we define first.

Roughness. On exact quotes the interpolating spline is the right choice and the penalty should be zero; on noisy quotes an interpolant transmits the noise to the second derivative. The rules read the roughness of the thriced interpolant, the estimator at $c = 0$, from a score that is one for a unimodal curve and larger otherwise. For a nonnegative function g on a window $[a, b]$, let

$$\mathrm{TV}(g) = \frac{g(a) + \int_a^b |g'| + g(b)}{2 \max g}, \tag{8}$$

the total variation of g completed to zero at the two endpoints, divided by twice its maximum. A unimodal curve scores exactly one; an additional mode raises the score by its height relative to the maximum. The score is applied to the positive part of the thriced interpolant on $[0.60F, 1.30F]$, a window that holds the bulk

of the mass on every chain we have seen and avoids the far wings, where the estimate is dominated by the tail completion. The thriced kernel itself, scored the same way, gives $\mathrm{TV}_* = 1.007$, slightly above one because of its sidelobes, and a nonnegative kernel of the same height has signed variation $v_* = \int |L_1| = 1.196$, so one excess unit of variation $v_* - 1 = 0.196$ is what one extra sidelobe of the kernel contributes. The roughness at bandwidth h is the excess of the interpolant over the allowance in those units,

$$\rho(h) \;=\; \frac{(\mathrm{TV} - \mathrm{TV}_*)_+}{v_* - 1}, \tag{9}$$

where TV is the score of the thriced interpolant convolved at h. On exact quotes of every regular design in Section 5 the score is below the allowance and $\rho = 0$; on the listed chains of Section 6 it is at most 0.05; on the noisy designs of Section 5 it is between two and fifteen, and it grows with the noise level. The constants 0.60, 1.30, and the choice of the kernel's own score as the allowance are conventions; Section 5.4 shows the estimator with the penalty switched off.

Bandwidth. A kernel estimator of a second derivative balances a squared bias of order h^4 (order two kernel) against a variance of order $(nh^5)^{-1}$, which gives $h \asymp n^{-1/9}$ (Fan and Gijbels, 1996); the familiar density rate $n^{-1/5}$ leaves nh^5 bounded and is not usable. The natural scale of a chain is $F\hat{\sigma}_{\mathrm{ATM}}\sqrt{T}$, the standard deviation of S_T under a lognormal at the at-the-money implied volatility $\hat{\sigma}_{\mathrm{ATM}}$, taken as the median implied volatility of contracts with $|K - F| \le 0.03F$. The base bandwidth is

$$h_0 \;=\; c_0\,\gamma\,F\,\hat{\sigma}_{\mathrm{ATM}}\,\sqrt{T}\,n^{-1/9}, \qquad c_0 = 0.34, \tag{10}$$

with n the number of quoted strikes, and $\gamma \in [0.90, 1.10]$ a correction for the shape of the distribution: the interquartile range of the positive second differences of the quoted call (the butterfly masses, a crude histogram of $f_{\mathbb{Q}}$) divided by the normal interquartile range $2\Phi^{-1}(3/4)F\hat{\sigma}_{\mathrm{ATM}}\sqrt{T}$, clipped to that interval. The factor γ widens the bandwidth on fat-tailed chains and narrows it on peaked ones; on the listed chains of Section 6 it is at its lower bound. The rate $n^{-1/9}$ carries no noise level, whereas the bandwidth that balances bias and variance grows with the noise, as a power of its variance with exponent between $1/13$ and $1/17$ for the smoothers at hand. The roughness (9) supplies the missing number: it is zero on exact quotes and grows with the noise, so the rule widens the base bandwidth by

$$h \;=\; h_0\,(1 + \rho(h_0))^{1/6}. \tag{11}$$

On the listed chains the factor is at most 1.01 and h is h_0; on the noisy simulation designs it is between 1.2 and 1.6. The constant $c_0 = 0.34$ was chosen by inspection on the exact Heston and variance-gamma designs and on the four listed chains of Table 11, and the exponent $1/6$ on the noisy designs of Section 5. Neither comes from an asymptotic formula, and Section 5.4 therefore reports the estimator at $c_0 \in \{0.20, 0.27, 0.34, 0.45, 0.60\}$: the hold-out error on the listed chains varies by less than one part in ten between 0.20 and 0.45 and the Monte Carlo ranking does not change. Any bounded random multiple of $n^{-1/9}$ satisfies the bandwidth condition of Theorem 1.

Penalty. The penalty is chosen so that the thriced spline attenuates the frequency $1/h$ by the factor $1/(1+\rho)$, with $\rho = \rho(h)$ read at the widened bandwidth: one unit of excess roughness, one sidelobe's worth, is damped like one unit of the kernel's own allowance. On a uniform mesh of spacing δ the thriced spline multiplies a Fourier mode of frequency ω by $1 - (x/(1+x))^3$ with $x = cF^3\delta\omega^4$, by (7), and solving for the attenuation $1/(1+\rho)$ at $\omega = 1/h$ gives

$$c \;=\; x(\rho)\,\frac{h^4}{\delta\,F^3}, \qquad x(\rho) = \frac{t}{1-t}, \quad t = \Big(\frac{\rho}{1+\rho}\Big)^{1/3}, \tag{12}$$

with δ the median knot spacing. For the plain smoothing spline the same calibration gives $x = \rho$; the thriced spline needs the larger x, between $\rho^{1/3}$ for small ρ and 3ρ for large ρ, because each pass undoes part of the damping of the one before. The interpolant is computed twice, at h_0 and at h, the score twice, and the penalized spline three times. The rule does not count modes: a bimodal target raises c through the score, but a second mode that survives the smoothing is kept, as it is on two of the long-dated chains in Appendix D. On exact quotes of every regular design the penalty is zero.

4. Asymptotics

Write P for the integrated-tail distribution function of Section 3.1, so that $f_{\mathbb{Q}} = -FP''$ on $(0,\infty)$, and let the estimator be the thriced kernel (6) applied to the thriced spline (7) with the tails of Section 3.3, the bandwidth (10)–(11), and the penalty (12).

Assumption 1. *(i)* $\mathbb{E}^{\mathbb{Q}}[S_T] < \infty$ *and* $f_{\mathbb{Q}}$ *is four times continuously differentiable on* $(0,\infty)$*, so that* $P \in C^6(0,\infty)$.

(ii) Fix a compact $[A, B] \subset (0, \infty)$ *that contains the window* $[0.60F, 1.30F]$ *of* (12) *in its interior, and on which* $f_{\mathbb{Q}}$ *is positive somewhere. The strike grids* $0 < K_1^{(m)} < \cdots < K_m^{(m)}$ *satisfy* $K_1^{(m)} \le A$, $K_m^{(m)} \ge B$, *and are quasi-uniform on* $[A, B]$*: with* $n = n_m$ *the number of strikes in* $[A, B]$ *and* $\bar{\delta} = \bar{\delta}_m$ *the largest gap there,* $n\bar{\delta}$ *is bounded above and below, and the ratio of the largest to the smallest gap on* $[A, B]$ *is bounded. The median knot spacing* δ *of* (12) *is then of order* $1/n$.

(iii) The observations are $P_i = P(K_i) + \eta_i$ *with* η_i *independent, mean zero, and sub-Gaussian with a common proxy:* $\mathbb{E}e^{t\eta_i} \le e^{t^2\tau^2/2}$ *for all real* t. *Noise bounded by the quoted half-spread satisfies this.*

(iv) F *is fixed, the base bandwidth* $h_0 = h_{0,m}$ *is deterministic with* $h_0 \to 0$, $\bar{\delta}/h_0 \to 0$, *and* $nh_0^5/\log n \to \infty$.

Theorem 1. *Under Assumption 1, the estimator* $\hat{f}_{\mathbb{Q}}^{\text{th}}$ *with the tails of Section 3.3, the widened bandwidth* (11) *and the penalty* (12) *converges to* $f_{\mathbb{Q}}$ *in probability, uniformly on every compact subset of* (A, B).

The proof is in Appendix A. It proceeds in six steps, the first five at a deterministic bandwidth h. At penalty zero the estimator is a kernel estimator of P'' applied to the interpolating spline; its deterministic part converges by the interpolation error of the natural cubic spline and the moment conditions on the thriced kernel, and its noise part is a sub-Gaussian process with pointwise variance of order $\tau^2/(nh^5)$ and a Lipschitz modulus of the same order on the scale h, so a maximal inequality over a net of $O(1/h)$ points gives uniform convergence at the rate $\sqrt{\log n/(nh^5)}$. The random penalty is then shown to be small: the total-variation score of the interpolant exceeds that of $f_{\mathbb{Q}}$ by at most the variation of the noise process, which is $O_p((nh^7)^{-1/2})$, so the roughness is $O_p(1)$ and the smoothing-spline half-width $\beta = (cF^3\delta)^{1/4}$ is $O_p(h + (h/n)^{1/8}) = o_p(1)$. The bias at any deterministic penalty of that size is $O(\beta^4)$ by the derivative error bounds for smoothing splines, which the three passes of (7) preserve, and the noise at a random penalty is controlled by a chaining argument over the pair (strike, log penalty), which costs a further $\sqrt{\log n}$; the thriced spline is a polynomial in the smoothing-spline operator with eigenvalues in $[0, 1]$, so the same argument applies to it. Finally the widened bandwidth (11) lies between h_0 and a fixed multiple of h_0 with probability tending to one, because the roughness at h_0 is $O_p(1)$, and the bounds hold uniformly over that range by one more chaining coordinate. The condition $nh_0^5/\log n \to \infty$ is what the chaining needs; the rule (10) is a bounded multiple of $n^{-1/9}$, so $nh_0^5 \asymp n^{4/9}$ and the condition holds, as does $\bar{\delta}/h_0 \asymp n^{-8/9} \to 0$.

Remark 1. *At penalty zero and under six continuous derivatives of $f_{\mathbb{Q}}$, Step 1 of the proof gives a bias of order $h^6 + \bar{\delta}^4 h^{-2}$, the second term from cubic spline interpolation of a C^4 function passed through the second derivative of the kernel, and Step 2 a standard deviation of order $(nh^5)^{-1/2}$; the pointwise rate at $h \asymp n^{-1/17}$ would then be $n^{-6/17}$, the rate of a sixth-order kernel estimator of a second derivative. The base rule* (10) *uses the second-order exponent $n^{-1/9}$, which is conservative for the variance, and the widening* (11) *moves the bandwidth toward the noise-dependent optimum without changing the rate; the simulation evidence in Section 5 shows that on chains of* 32 *to* 256 *strikes the rule brings the estimator within a factor of two of its oracle error. A formal rate and a limiting distribution for the penalized estimator are left for future work.*

Remark 2. *The empirical section fits the spline to the decreasing-convex projection of the quoted calls rather than to the raw quotes. When the true call lies in the interior of the cone of decreasing convex functions, the projection changes the fitted values by an amount of the same stochastic order as η_i, and the theorem applies unchanged. The theorem does not require the estimator to be a density, and $\hat{f}^{\text{th}}_{\mathbb{Q}}$ is not one in finite samples; the negative mass vanishes in the limit with the sidelobes.*

5. Simulation Evidence

5.1. Designs

Three data-generating processes are used. The first is the stochastic-volatility model of Heston (1993) with the equity parameters of Bakshi et al. (1997): $S_0 = 100$, $r = 0.05$, $q = 0$, $T = 1/4$, mean reversion $\kappa_v = 1.15$, long-run variance $\theta_v = 0.04$, volatility of variance $\sigma_v = 0.39$, correlation $\rho_v = -0.64$, and $v_0 = 0.04$. The second is the variance-gamma process of Madan et al. (1998) with the parameters of Carr and Madan (1999): $\sigma = 0.12$, $\nu = 0.2$, $\theta = -0.14$, a pure-jump Lévy process whose density of S_T is more peaked than Heston's. The third is the jump diffusion of Bates (1996) with the SVJ estimates of Bakshi et al. (1997): $\kappa_v = 2.03$, $\theta_v = 0.04$, $\sigma_v = 0.38$, $\rho_v = -0.52$, jump intensity $\lambda = 0.59$, mean log jump $\mu_J = -0.05$, and jump volatility $\sigma_J = 0.07$. Call prices are computed from the characteristic function by the damped integrand of Carr and Madan (1999) on 4096 frequencies in $(0, 250]$, and the true density of S_T is the Fourier inversion of the same characteristic function on the same grid; for variance gamma the density is the Bessel-function formula of Madan et al. (1998) and the calls are discounted payoffs under it.

Three strike meshes are used. The dense chain has $m = 256$ equally spaced strikes on $[30, 220]$ (spacing 0.745), the sparse chain $m = 32$ equally spaced strikes on $[70, 140]$ (spacing 2.26), and the listed chain is the S&P 500 December mesh of Section 6 rescaled to a forward of 100 and thinned to 110 strikes, so that it is fine near the money and coarse in the wings. Integrated squared error (ISE) is $\int_{60}^{150} (\hat{f}_{\mathbb{Q}} - f_{\mathbb{Q}})^2 \, \mathrm{d}K$, evaluated by the trapezoidal rule on 401 points, and is reported multiplied by 10^4 in the Monte Carlo tables.

5.2. Exact quotes

Table 1 reports, for the four nested estimators—the quoted spline, the spline with tails, tails with twicing, and tails with thricing—the ISE at the oracle bandwidth h^*, which minimizes the ISE on a geometric grid, and at the rule bandwidth (10). On these exact quotes the penalty rule selects zero.

Table 1: Integrated squared error on exact quotes for the nested estimators. h^* minimizes the ISE on a geometric grid; $h_{1/9}$ is the rule (10), with the penalty at zero as the rule selects. Quoted spline uses only the observed strikes; Tails adds the wings of Section 3.3; twice and thrice are the two- and three-bandwidth jackknives of Section 3.4.

Design	Estimator	h^*	Oracle ISE	$h_{1/9}$	Rule ISE
Heston, dense	Quoted spline	0.15	9.76×10^{-10}	1.75	1.01×10^{-5}
	Tails	0.15	9.76×10^{-10}	1.75	1.01×10^{-5}
	Tails + twice	0.39	2.68×10^{-12}	1.75	9.49×10^{-8}
	Tails + thrice	0.63	1.16×10^{-12}	1.75	6.85×10^{-9}
Heston, sparse	Quoted spline	16.19	1.80×10^{-2}	2.20	1.28×10^{-1}
	Tails	0.45	1.13×10^{-6}	2.20	2.47×10^{-5}
	Tails + twice	1.64	9.67×10^{-7}	2.20	1.29×10^{-6}
	Tails + thrice	2.18	9.19×10^{-7}	2.20	9.20×10^{-7}
VG, dense	Quoted spline	0.17	5.12×10^{-7}	1.03	7.94×10^{-5}
	Tails	0.17	5.12×10^{-7}	1.03	7.94×10^{-5}
	Tails + twice	0.33	3.93×10^{-7}	1.03	1.43×10^{-5}
	Tails + thrice	0.46	4.04×10^{-7}	1.03	8.51×10^{-6}
VG, sparse	Quoted spline	14.03	3.27×10^{-2}	1.30	2.17×10^{-1}
	Tails	0.45	3.97×10^{-5}	1.30	1.77×10^{-4}
	Tails + twice	0.80	3.31×10^{-5}	1.30	5.06×10^{-5}
	Tails + thrice	0.92	3.13×10^{-5}	1.30	3.76×10^{-5}
Bates, dense	Quoted spline	0.15	5.38×10^{-10}	1.86	7.31×10^{-6}
	Tails	0.15	5.38×10^{-10}	1.86	7.31×10^{-6}
	Tails + twice	0.39	3.24×10^{-13}	1.86	3.80×10^{-8}
	Tails + thrice	0.54	5.82×10^{-15}	1.86	1.49×10^{-9}
Bates, sparse	Quoted spline	16.19	1.69×10^{-2}	2.34	1.20×10^{-1}
	Tails	0.60	1.15×10^{-6}	2.34	1.81×10^{-5}
	Tails + twice	1.89	9.40×10^{-7}	2.34	1.02×10^{-6}
	Tails + thrice	2.52	8.71×10^{-7}	2.34	8.86×10^{-7}

Three regularities appear on every target. On the dense chain the first strike lies far below the window and the last call is already zero, so the wings do not change the estimate; twicing lowers the rule-bandwidth error by one to two orders of magnitude and thricing by a further factor of two to fourteen. The rule bandwidth is wider than the oracle on the dense chain, and the error at the oracle is correspondingly smaller. On the sparse chain the quoted spline without tails misses the window entirely, because the natural boundary condition is imposed at $K = 70$ and $K = 140$; the left line of Section 3.3 restores it, and each jackknife step then reduces the error further. On the sparse chain thricing at the rule

bandwidth is close to its oracle. Figures 1 and 2 plot the four estimates at the rule bandwidth on the Heston and variance-gamma targets.

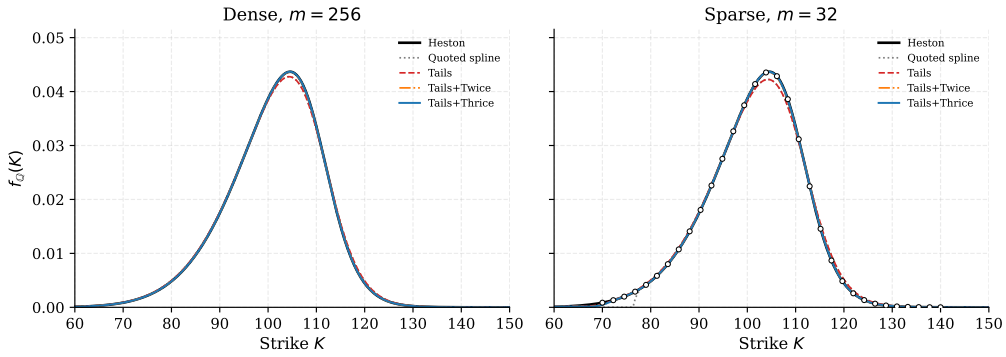


Figure 1: Heston density of S_T on $[60, 150]$. Quoted spline, tails, tails plus twice, and tails plus thrice, each at the rule bandwidth (10). Sparse quoted strikes are marked on the true density. Curves are truncated at zero.

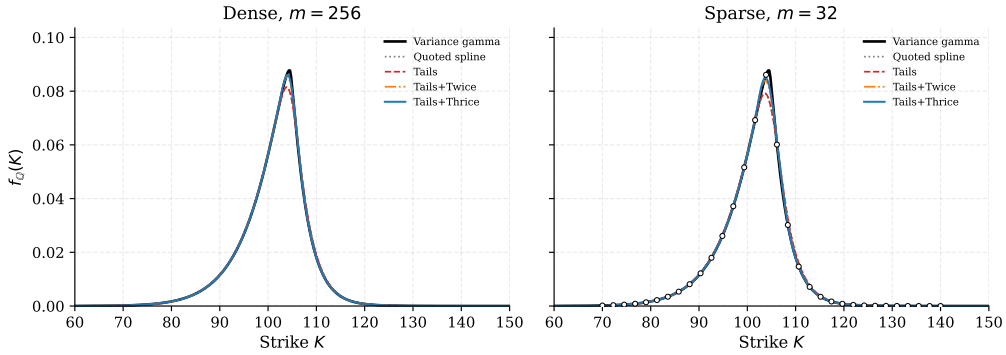


Figure 2: Variance-gamma density of S_T on $[60, 150]$. Layout as in Figure 1.

Table 2 scores the estimator against the methods of Section 6.2 on the same exact chains, each tuned as described there. Exact quotes favour interpolation, and the cross-validated competitors select the smallest bandwidth on their grid; the table is reported for completeness, and the noisy designs of the next section are the informative ones.

Table 2: Integrated squared error on exact quotes, $[60, 150]$. Every method is tuned as in Section 6.2; the first block is the common cross-validation, the second block the rules of the source papers. The column minimum is in bold.

	Heston		VG		Bates	
Estimator	dense	sparse	dense	sparse	dense	sparse
Ours, rule	6.85×10^{-9}	9.20×10^{-7}	**8.51e-6**	**3.76e-5**	1.49×10^{-9}	8.86×10^{-7}
Ours, CV	**3.86e-9**	1.07×10^{-5}	2.31×10^{-5}	6.39×10^{-4}	**5.98e-11**	1.35×10^{-6}
PCA, CV	1.56×10^{-5}	**4.93e-7**	2.32×10^{-4}	3.13×10^{-4}	1.04×10^{-10}	**5.29e-8**
AS–Lo, CV	5.78×10^{-8}	4.82×10^{-6}	6.58×10^{-5}	4.82×10^{-4}	1.95×10^{-8}	8.55×10^{-7}
AS–Duarte, CV	3.12×10^{-6}	2.44×10^{-4}	1.76×10^{-4}	2.20×10^{-3}	2.18×10^{-6}	7.80×10^{-5}
GHS IV, CV	3.98×10^{-9}	2.03×10^{-6}	3.27×10^{-5}	3.68×10^{-4}	9.87×10^{-10}	2.57×10^{-7}
GHS call, CV	3.19×10^{-6}	2.81×10^{-4}	1.87×10^{-4}	2.38×10^{-3}	2.25×10^{-6}	9.06×10^{-5}
AS–Lo, App. A	1.11×10^{-3}	1.09×10^{-3}	1.05×10^{-2}	6.78×10^{-3}	6.92×10^{-4}	6.10×10^{-4}
AS–Duarte, plug-in	6.13×10^{-4}	2.32×10^{-5}	6.47×10^{-3}	1.13×10^{-3}	3.91×10^{-4}	1.24×10^{-5}
GHS IV, LOO	6.24×10^{-9}	5.66×10^{-7}	2.16×10^{-5}	2.53×10^{-4}	1.25×10^{-9}	1.85×10^{-7}
GHS IV, 2×LOO	9.27×10^{-8}	4.34×10^{-6}	1.15×10^{-4}	7.03×10^{-4}	1.91×10^{-8}	1.11×10^{-6}
GHS call, LOO	7.52×10^{-7}	5.59×10^{-5}	5.39×10^{-5}	1.36×10^{-3}	4.20×10^{-7}	3.31×10^{-5}
GHS call, 2×LOO	1.14×10^{-5}	6.32×10^{-4}	4.45×10^{-4}	6.25×10^{-3}	6.49×10^{-6}	4.27×10^{-4}

5.3. Noisy quotes

Noise is added in implied volatility: each quote receives an independent $N(0, \mathrm{sd}^2)$ perturbation of its Black–Scholes implied volatility, with $\mathrm{sd} \in \{0.005, 0.01, 0.02\}$, values clipped to $[0.02, 2.5]$, and the call rebuilt. Every replication starts from the decreasing-convex projection of the noisy calls, and every method, including ours, is tuned on that draw as described in Section 6.2. Table 3 reports the mean ISE over 200 replications at $\mathrm{sd} = 0.01$, with Monte Carlo standard errors in parentheses, for the three targets and three meshes; Table 4 repeats the dense and sparse designs at the two other noise levels with 100 replications. In each column every entry within two standard errors of the column minimum is in bold.

Table 3: Mean integrated squared error ($\times 10^4$) on noisy quotes, implied-volatility noise sd = 0.01, 200 replications, Monte Carlo standard error in parentheses. The first block is the common cross-validation of Section 6.2; the second block is the rule of each source paper; the third block is each competitor at our rule bandwidth (11) (Section 5.6). Entries within two standard errors of the column minimum are in bold.

	Heston			VG			Bates		
Estimator	dense	sparse	listed	dense	sparse	listed	dense	sparse	listed
Ours, rule	**1.50** (0.08)	**2.72** (0.13)	**2.11** (0.08)	12.11 (0.47)	19.58 (0.82)	**12.73** (0.39)	**1.18** (0.09)	**1.94** (0.10)	**1.26** (0.06)
Ours, CV	5.60 (0.30)	3.40 (0.18)	46.90 (2.01)	13.04 (0.63)	19.58 (0.63)	42.54 (2.19)	5.63 (0.37)	2.84 (0.14)	38.16 (1.68)
PCA, CV	3.96 (0.24)	4.44 (0.30)	16.20 (1.80)	**8.40** (0.33)	13.61 (0.51)	33.36 (5.30)	3.71 (0.23)	4.29 (0.33)	13.93 (1.96)
AS–Lo, CV	81.98 (2.27)	4.95 (0.29)	3500 (201)	79.54 (3.29)	**9.77** (0.42)	788 (40)	76.99 (2.22)	5.40 (0.29)	5347 (332)
AS–Duarte, CV	77.11 (2.13)	5.67 (0.24)	329 (14)	71.74 (2.96)	29.72 (0.65)	317 (10)	72.82 (2.10)	5.37 (0.22)	278 (7.03)
GHS IV, CV	180 (5.23)	319 (72)	261 (12)	199 (8.22)	606 (202)	513 (16)	183 (5.60)	531 (136)	258 (11)
GHS call, CV	68.99 (2.03)	11.22 (0.50)	25.82 (1.93)	61.95 (2.71)	34.01 (0.67)	14.29 (0.47)	64.02 (1.92)	9.92 (0.52)	51.04 (5.53)
AS–Lo, App. A	15.96 (0.04)	11.07 (0.02)	12.12 (0.02)	100 (0.19)	69.12 (0.15)	63.39 (0.08)	8.18 (0.03)	6.16 (0.01)	6.58 (0.02)
AS–Duarte, plug-in	6.27 (0.07)	37.07 (1.24)	57.10 (1.38)	64.70 (0.21)	24.33 (1.02)	29.55 (0.92)	4.05 (0.06)	42.01 (1.35)	62.99 (1.57)
GHS IV, LOO	177 (4.12)	65.94 (32)	188 (4.10)	200 (6.38)	299 (179)	183 (5.14)	176 (4.94)	534 (184)	174 (4.20)
GHS call, LOO	170 (3.76)	13.97 (0.60)	194 (3.77)	181 (5.61)	22.83 (0.84)	160 (4.46)	177 (3.93)	13.91 (0.57)	185 (3.91)
PCA, at (11)	14.11 (0.88)	11.36 (0.65)	8.62 (0.43)	61.42 (3.19)	39.41 (2.89)	53.48 (6.54)	10.56 (0.62)	9.96 (0.58)	7.04 (0.37)
AS–Lo, at (11)	9.62 (0.46)	12.42 (0.81)	6.67 (0.25)	51.96 (2.53)	70.74 (3.61)	23.17 (0.95)	8.64 (0.49)	9.89 (0.53)	5.54 (0.24)
AS–Duarte, at (11)	8.66 (0.39)	10.96 (0.66)	14.85 (0.32)	46.78 (2.22)	62.45 (3.02)	32.64 (1.08)	7.75 (0.42)	8.70 (0.44)	13.11 (0.31)
GHS IV, at (11)	21.03 (0.89)	29.00 (1.55)	27.68 (1.20)	95.49 (4.20)	153 (5.95)	55.28 (1.98)	18.45 (0.87)	24.92 (1.15)	23.60 (1.01)
GHS call, at (11)	16.34 (0.74)	20.57 (1.19)	27.39 (1.37)	79.32 (3.62)	142 (6.33)	53.61 (2.02)	14.53 (0.74)	17.30 (0.84)	21.17 (0.90)

Table 4: Mean integrated squared error ($\times 10^4$) at sd = 0.005 (upper panel) and sd = 0.02 (lower panel), dense and sparse chains, 100 replications. Layout as in Table 3.

	Heston		VG		Bates	
Estimator	dense	sparse	dense	sparse	dense	sparse
sd = 0.005						
Ours, rule	**0.67** (0.05)	1.26 (0.11)	7.71 (0.34)	11.53 (0.55)	**0.51** (0.04)	**1.08** (0.08)
Ours, CV	1.70 (0.11)	1.70 (0.13)	**7.06** (0.37)	11.27 (0.51)	1.41 (0.09)	1.78 (0.17)
PCA, CV	1.50 (0.15)	2.34 (0.23)	**6.59** (0.32)	12.60 (0.77)	1.38 (0.12)	2.00 (0.28)
AS–Lo, CV	35.71 (1.46)	**0.84** (0.09)	31.39 (1.93)	**6.43** (0.27)	30.65 (1.39)	1.69 (0.16)
AS–Duarte, CV	33.87 (1.38)	2.83 (0.10)	29.72 (1.80)	27.11 (0.59)	29.16 (1.32)	2.46 (0.11)
GHS IV, CV	104 (4.53)	383 (151)	89.91 (5.71)	479 (231)	88.94 (4.48)	396 (243)
GHS call, CV	28.50 (1.26)	7.50 (0.48)	24.06 (1.55)	30.53 (0.60)	24.58 (1.20)	6.30 (0.37)
AS–Lo, App. A	15.53 (0.04)	10.93 (0.01)	96.48 (0.25)	68.16 (0.09)	7.92 (0.03)	6.13 (0.01)
AS–Duarte, plug-in	6.18 (0.06)	15.51 (0.68)	64.59 (0.16)	14.78 (0.72)	3.92 (0.04)	19.60 (0.98)
GHS IV, LOO	90.39 (3.85)	68.23 (34)	97.12 (4.26)	148 (73)	89.49 (3.83)	223 (143)
GHS call, LOO	90.41 (3.03)	5.89 (0.37)	87.47 (4.02)	16.35 (0.67)	89.79 (3.18)	5.93 (0.32)
PCA, at (11)	8.99 (0.57)	5.49 (0.39)	41.26 (4.87)	30.66 (3.80)	7.43 (0.68)	6.72 (0.57)
AS–Lo, at (11)	3.00 (0.17)	3.87 (0.28)	19.70 (1.46)	28.07 (2.83)	2.35 (0.14)	4.10 (0.30)
AS–Duarte, at (11)	2.97 (0.15)	4.03 (0.23)	19.45 (1.34)	27.11 (2.46)	2.33 (0.12)	3.94 (0.25)
GHS IV, at (11)	6.91 (0.34)	9.93 (0.57)	39.10 (2.45)	71.47 (4.14)	5.63 (0.29)	10.13 (0.62)
GHS call, at (11)	5.52 (0.30)	7.61 (0.44)	33.75 (2.21)	76.26 (5.93)	4.49 (0.25)	7.86 (0.51)
sd = 0.02						
Ours, rule	**3.28** (0.29)	**5.39** (0.69)	30.62 (3.18)	53.13 (5.43)	**2.50** (0.23)	**4.68** (0.50)
Ours, CV	19.04 (1.40)	**5.90** (0.48)	36.92 (3.54)	31.63 (1.31)	20.25 (2.19)	**5.59** (0.42)
PCA, CV	7.68 (0.59)	9.43 (1.01)	**13.64** (0.88)	**23.28** (1.46)	7.38 (0.51)	10.72 (0.94)
AS–Lo, CV	150 (5.43)	14.99 (1.29)	177 (10)	**24.12** (1.80)	142 (5.93)	16.32 (1.04)
AS–Duarte, CV	140 (5.05)	12.42 (0.98)	159 (9.15)	37.90 (1.77)	134 (5.61)	13.03 (0.80)
GHS IV, CV	300 (12)	623 (216)	379 (20)	2788 (1193)	269 (13)	720 (290)
GHS call, CV	129 (4.86)	17.78 (1.32)	143 (8.61)	40.43 (1.59)	122 (5.41)	19.52 (1.32)
AS–Lo, App. A	17.39 (0.10)	11.50 (0.05)	109 (0.33)	72.90 (0.38)	9.20 (0.07)	6.47 (0.04)
AS–Duarte, plug-in	6.44 (0.20)	71.90 (3.39)	65.05 (0.58)	51.18 (3.50)	4.22 (0.18)	77.68 (3.28)
GHS IV, LOO	289 (9.48)	260 (195)	369 (16)	1076 (761)	286 (9.05)	412 (351)
GHS call, LOO	273 (8.46)	31.70 (1.91)	332 (14)	43.71 (2.85)	271 (8.35)	33.45 (1.96)
PCA, at (11)	22.88 (1.73)	18.64 (1.50)	98.46 (5.96)	66.18 (7.52)	18.77 (1.52)	17.85 (1.44)
AS–Lo, at (11)	26.41 (1.90)	31.20 (2.80)	132 (10)	193 (17)	22.80 (1.94)	27.75 (2.38)
AS–Duarte, at (11)	22.59 (1.62)	25.48 (2.32)	117 (9.07)	164 (14)	19.47 (1.67)	22.83 (1.96)
GHS IV, at (11)	52.06 (3.29)	613 (537)	218 (15)	265 (16)	44.13 (3.16)	72.79 (9.76)
GHS call, at (11)	40.11 (2.74)	47.02 (3.78)	183 (13)	263 (15)	34.38 (2.70)	43.03 (3.37)

Table 5 separates the estimator from its tuning. For each method and each replication it reports the ISE at the infeasible oracle bandwidth, the point of the

common grid that minimizes the ISE on that draw; our estimator keeps the penalty rule at every candidate. The oracle rows bound what any tuning rule could achieve.

Table 5 is the most informative table of the section, and it is not flattering to our kernel alone. At the oracle bandwidth the implied-volatility smoothers are the most accurate on the dense and sparse meshes: the local cubic of implied volatility has an ISE of 0.33 on dense Heston against 1.00 for ours, and the implied-volatility kernel 0.51 against 2.26 on sparse Heston; PCA is also ahead of ours on most regular designs. The reason is the transform: implied volatility is nearly flat in strike, so a smoother of implied volatility has a bias that is tiny at bandwidths where a smoother of P does not, and the sixth-order kernel's extra variance is then a cost without a matching benefit. On the listed-shaped mesh the picture changes, because wide gaps in the wings break the local polynomial windows and the implied-volatility kernel: ours and PCA are tied on Heston, and ours is within a factor of two of the best on the other two targets. The feasible rows of Table 3 should be read against these oracles. Our rule reaches within a factor of 1.0 to 2.0 of our oracle on every design, and on sparse variance gamma it is at the oracle. The published rules of the two implied-volatility smoothers are off their oracles by factors of forty to five hundred on the dense mesh, and their price-cross-validated versions by more; the plug-in of Aït-Sahalia and Duarte is within a factor of three of its oracle on the dense mesh, but that oracle is above our rule. The practical advantage documented in this paper is therefore that of a procedure whose tuning is settled, and Section 5.6 asks how much of it transfers when the competitors are handed the same bandwidth rule.

Results. At sd = 0.01 the rule estimator has the smallest mean ISE, or one within two standard errors of the smallest, in seven of the nine designs of Table 3: all three meshes on Heston and on Bates, and the listed mesh on variance gamma. Its advantage is largest on the listed mesh, which is the most realistic of the three, where the next method is PCA at eight to eleven times the error on Heston and Bates. On the dense mesh the margin over PCA is a factor of 2.6 on Heston and 3.1 on Bates, and on the sparse mesh 1.6 and 2.2. The two exceptions are both variance gamma, whose density is much more peaked than the other two: on the dense mesh PCA is better by a factor of 1.4, and on the sparse mesh the cross-validated implied-volatility kernel of Aït-Sahalia and Lo is better by a factor of two. Table 7 shows that no bandwidth constant closes that gap, so the peak is where the sixth-order kernel's negative sidelobes cost most; the variance-gamma density at a quarter year has a cusp at the forward whose second derivative is unbounded, which no global-bandwidth linear smoother resolves.

Table 5: Mean integrated squared error ($\times 10^4$) at the oracle bandwidth, sd = 0.01, one hundred replications, standard error in parentheses. The oracle bandwidth minimizes the ISE of the draw on the common grid and is infeasible. Entries within two standard errors of the column minimum are in bold.

Estimator	Heston dense	Heston sparse	Heston listed	VG dense	VG sparse	VG listed	Bates dense	Bates sparse	Bates listed
Ours	1.00 (0.07)	2.26 (0.16)	**1.04** (0.07)	10.08 (0.53)	19.16 (0.93)	9.76 (0.40)	0.78 (0.05)	1.32 (0.11)	0.68 (0.04)
PCA	0.69 (0.05)	1.44 (0.09)	**0.93** (0.07)	5.60 (0.28)	11.11 (0.53)	5.51 (0.27)	0.36 (0.02)	0.77 (0.05)	**0.47** (0.04)
AS–Lo	0.37 (0.02)	**0.51** (0.03)	2.00 (0.09)	5.59 (0.28)	**8.98** (0.52)	8.50 (0.38)	0.18 (0.01)	**0.35** (0.03)	1.32 (0.06)
AS–Duarte	2.72 (0.11)	4.68 (0.23)	12.09 (0.33)	14.70 (0.66)	30.34 (0.97)	27.53 (1.00)	2.38 (0.09)	3.14 (0.17)	11.69 (0.36)
GHS IV	**0.33** (0.02)	1.97 (0.38)	1.59 (0.20)	**4.92** (0.23)	12.84 (1.14)	**4.81** (0.21)	**0.14** (0.01)	1.45 (0.44)	3.76 (0.65)
GHS call	2.35 (0.11)	7.78 (0.36)	7.97 (0.64)	13.59 (0.65)	21.23 (1.18)	13.46 (0.56)	2.10 (0.09)	6.72 (0.34)	11.59 (0.89)

The cross-validated kernel smoothers, the implied-volatility kernel, the local linear and the two local cubics, are one to three orders of magnitude worse than the rule on the dense and listed meshes. Their price cross-validation selects the grid floor, as on the listed chains, and the resulting second derivative is noise; on the sparse mesh, where the grid floor is wider relative to the strike range, the same methods are competitive, and the implied-volatility kernel is within a factor of 1.8 of the rule on Heston. The source papers' own rules do better than their cross-validated versions on the dense mesh and worse on the sparse one, the plug-in of Aït-Sahalia and Duarte being the most accurate of them on dense chains. "Ours, CV" is also affected: it is one to four times worse than the rule on the dense and sparse meshes and three to thirty times worse on the listed mesh. The rule bandwidth does not suffer from this, because it is tied to the scale of the distribution, to $n^{-1/9}$ and to the roughness, rather than to a pricing criterion that cannot see the second derivative.

Table 4 varies the noise. At sd = 0.005 the dense-mesh ranking is unchanged on Heston and Bates and PCA edges ours on variance gamma, and on the sparse mesh the implied-volatility kernel is best on Heston and variance gamma, because with little noise and few strikes the cross-validation is close to interpolation of a smooth smile; ours is best on sparse Bates. At sd = 0.02 the rule estimator is best or tied on all four Heston and Bates designs, by a factor of two over the next method on the dense meshes, and PCA is best on variance gamma, with a margin of 2.3. The roughness rule is what carries the estimator across the noise levels: the mean bandwidth of Table 6 is 1.3 to 1.5 times the base bandwidth (10) at sd = 0.01, and the widening grows with the noise, so that the fixed constant c_0 does not have to be right for every noise level. The diagnostics also show that the penalty rule is active on every noisy replication, and that the negative mass of the thriced estimate is below 0.01 on average and at most 0.03 on any draw, so that clipping at zero and renormalizing changes the ISE in the third digit (Section 5.5).

Table 6 records, for the rule estimator at sd = 0.01, the mean widened bandwidth (11), the share of replications on which the penalty rule was positive, and the negative mass $\int \hat{f}_{\mathbb{Q}}^{-}$ on the evaluation window.

Table 6: Diagnostics of the rule estimator at sd = 0.01: mean widened bandwidth (11), share of replications with a positive penalty, and mean and maximum negative mass $\int_{60}^{150} \max(-\hat{f}^{\mathrm{th}}_{\mathbb{Q}}, 0)$.

Design	mean h	$c > 0$, %	mean neg. mass	max neg. mass
Heston, dense	2.52	100	0.0011	0.0070
Heston, sparse	3.09	100	0.0050	0.0151
Heston, listed	2.66	100	0.0037	0.0319
VG, dense	1.56	100	0.0015	0.0133
VG, sparse	1.87	100	0.0058	0.0239
VG, listed	1.68	100	0.0097	0.0304
Bates, dense	2.68	100	0.0002	0.0030
Bates, sparse	3.22	100	0.0019	0.0128
Bates, listed	2.83	100	0.0024	0.0314

5.4. Sensitivity to the rules

Table 7 reports the estimator at bandwidth constants $c_0 \in \{0.20, 0.27, 0.34, 0.45, 0.60\}$ in (10), each widened by (11), with the penalty rule (12) and with the penalty fixed at zero. The listed panel gives the nested hold-out error in half-spread units on the four chains of Section 6 and the number of modes of the fitted density; the simulation panel gives the mean ISE at sd = 0.01 over one hundred replications.

Table 7: Sensitivity to the bandwidth constant c_0 and to the penalty. Listed panel: nested even/odd hold-out error in units of the quoted half-spread, and number of modes of the fitted density on $[0.60F, 1.30F]$. Simulation panel: mean ISE ($\times 10^4$) at sd = 0.01, one hundred replications. "rule" is (12); "zero" is the interpolating spline.

		Hold-out, half-spreads					Modes				
Chain	Penalty	0.20	0.27	0.34	0.45	0.60	0.20	0.27	0.34	0.45	0.60
SPX Dec 2026	rule	0.107	0.105	0.106	0.108	0.165	1	1	1	1	1
	zero	0.108	0.105	0.105	0.108	0.165	1	1	1	1	1
SPX Mar 2027	rule	0.079	0.080	0.083	0.108	0.270	1	1	1	1	1
	zero	0.079	0.080	0.083	0.108	0.270	2	1	1	1	1
NDX Dec 2026	rule	0.076	0.076	0.076	0.079	0.086	1	1	1	1	1
	zero	0.075	0.076	0.077	0.079	0.086	8	4	3	1	1
RUT Dec 2026	rule	0.069	0.069	0.068	0.073	0.135	1	1	1	1	1
	zero	0.069	0.068	0.068	0.073	0.135	2	1	1	1	1
Simulation, mean ISE $\times 10^4$		c_0									
Design	Penalty	0.20	0.27	0.34	0.45	0.60					
Heston, dense	rule	2.73	1.59	1.38	1.35	1.48					
	zero	328.00	173.23	93.61	36.59	11.53					
Heston, sparse	rule	5.33	3.20	2.88	2.89	3.09					
	zero	376.52	198.06	103.56	40.51	13.19					
Heston, listed	rule	2.53	2.11	1.99	1.79	1.76					
	zero	246.59	125.21	65.21	24.63	7.57					
VG, dense	rule	27.28	15.37	12.89	12.75	12.83					
	zero	944.56	537.05	331.15	167.48	71.18					
VG, sparse	rule	46.26	21.75	18.25	18.48	18.45					
	zero	646.25	497.82	342.55	172.87	68.73					
VG, listed	rule	15.24	11.85	12.92	14.11	13.43					
	zero	626.59	356.82	209.96	94.15	33.52					
Bates, dense	rule	3.04	1.54	1.20	1.14	1.44					
	zero	303.65	164.23	91.17	37.07	12.06					
Bates, sparse	rule	3.69	2.26	2.06	2.05	2.42					
	zero	337.66	176.03	91.70	34.92	10.50					
Bates, listed	rule	1.71	1.38	1.33	1.23	1.46					
	zero	214.48	109.62	58.34	23.18	7.40					

Two conclusions follow from Table 7. The bandwidth constant hardly matters

on the listed chains between 0.20 and 0.34: the hold-out moves by a few thousandths of a half-spread, and the density keeps one mode at every constant once the penalty is on. At 0.45 the longer SPX chain starts to lose, and at 0.60 the bandwidth is wide enough to bias the far wings of every chain. In the simulations the ISE at $c_0 = 0.34$ is within a factor of 1.15 of the best constant on every design, including variance gamma, and between 0.27 and 0.60 it moves by less than a factor of 1.3; the constant 0.20 is clearly too narrow at this noise level. The flatness is the work of the widening (11), which reads the noise from the chain and takes the constant part of the way to where it should be. The penalty does the rest on noisy chains: with the penalty switched off the ISE is sixteen to eighty times larger at $c_0 = 0.34$, because the interpolating spline transmits the noise to the second derivative, and on NDX the interpolant has eight modes at the narrowest constant. On the listed chains the penalty barely changes the pricing error, which is the point of Section 6.3: pricing does not see the roughness that the density shows.

5.5. Variants of the estimator

Three modifications were tried and are reported for completeness in Table 8. The first replaces the penalty rule (12) by generalized cross-validation of the smoothing spline (Wahba, 1977; Green and Silverman, 1994), with the trace of the smoother estimated by Hutchinson's method; this is the textbook choice and removes the window and allowance constants of Section 3.5. The second clips the thriced density at zero and renormalizes it to unit mass, which makes the estimate a density. The third weights the observations in (3) by the inverse squared half-spread, the generalized-least-squares choice when the noise is proportional to the spread; it is evaluated on the listed chains, where spreads are observed, and under a heteroskedastic design in which the out-of-the-money premium receives noise with standard deviation $0.01 + 0.02\times$premium and the weights are the true inverse variances.

Table 8: Variants of the estimator. Simulation panel: mean ISE ($\times 10^4$) over one hundred replications at sd = 0.01 (implied-volatility noise) and under the heteroskedastic price-noise design; "+" is the positive-part renormalized estimate, "w" the inverse-variance weighted spline. Listed panel: nested hold-out in half-spreads and number of modes.

Implied-volatility noise				
Design	rule	rule+	GCV	GCV+
Heston, dense	1.47 (0.10)	**1.47 (0.10)**	107.43 (4.85)	103.35 (4.46)
Heston, sparse	2.82 (0.21)	**2.80 (0.21)**	89.39 (6.82)	84.02 (6.03)
Heston, listed	2.05 (0.13)	**1.92 (0.11)**	63.75 (3.11)	60.92 (2.89)
VG, dense	12.54 (0.66)	**12.52 (0.66)**	349.83 (17.04)	301.68 (12.98)
VG, sparse	20.25 (1.86)	**20.09 (1.81)**	229.27 (23.10)	174.39 (13.82)
VG, listed	12.87 (0.62)	**12.82 (0.62)**	172.96 (7.79)	160.00 (6.87)
Bates, dense	1.22 (0.10)	**1.22 (0.10)**	77.84 (4.10)	75.75 (3.81)
Bates, sparse	1.79 (0.14)	**1.78 (0.14)**	78.77 (5.85)	74.54 (5.18)
Bates, listed	1.35 (0.09)	**1.32 (0.09)**	61.12 (2.72)	56.56 (2.40)

Heteroskedastic price noise						
Design	rule	rule+	GCV	rule, w	rule, w+	GCV, w
Heston, dense	0.46 (0.03)	**0.46 (0.03)**	21.26 (1.46)	2.95 (0.13)	2.95 (0.13)	12.55 (1.14)
Heston, sparse	0.79 (0.06)	**0.79 (0.06)**	13.94 (1.59)	3.02 (0.17)	3.01 (0.17)	5.97 (0.99)
Heston, listed	0.74 (0.04)	**0.72 (0.03)**	10.35 (0.63)	1.40 (0.07)	1.37 (0.07)	10.35 (0.63)
VG, dense	**4.12 (0.17)**	4.12 (0.17)	60.46 (3.28)	15.64 (0.44)	15.66 (0.44)	22.79 (1.68)
VG, sparse	**6.37 (0.26)**	6.38 (0.26)	29.62 (2.30)	18.93 (0.53)	18.93 (0.53)	13.85 (0.87)
VG, listed	4.08 (0.14)	**4.07 (0.14)**	27.69 (1.35)	8.92 (0.21)	8.91 (0.21)	27.69 (1.35)
Bates, dense	0.42 (0.04)	**0.42 (0.04)**	17.65 (0.90)	2.67 (0.11)	2.67 (0.11)	12.54 (0.85)
Bates, sparse	0.65 (0.04)	**0.65 (0.04)**	10.15 (1.11)	2.89 (0.15)	2.88 (0.15)	4.05 (0.33)
Bates, listed	0.35 (0.02)	**0.34 (0.02)**	10.77 (0.50)	1.10 (0.03)	1.10 (0.03)	10.77 (0.50)

Listed chains								
	Hold-out, half-spreads				Modes			
Chain	rule	GCV	rule, w	GCV, w	rule	GCV	rule, w	GCV, w
SPX Dec 2026	0.106	0.105	0.108	0.105	1	1	1	1
SPX Mar 2027	0.083	0.083	0.083	0.083	1	1	1	1
NDX Dec 2026	0.076	0.077	0.076	0.077	1	3	1	3
RUT Dec 2026	0.068	0.068	0.068	0.069	1	1	1	1

None of the three variants improves on the estimator as published. Generalized cross-validation selects a penalty close to zero on every noisy design and the ISE is ten to seventy times that of the rule: GCV targets the fit to P, for which the interpolant is nearly optimal, and is blind to the roughness of the second derivative, which is the quantity of interest. On the listed chains GCV prices as well

as the rule but gives the NDX density three modes. The positive-part correction changes the ISE in the second or third digit, because the negative mass of the thriced estimate is already small; it is a free way of reporting a proper density and we recommend it for that purpose, but the tables of this paper report the uncorrected estimate so that the negative mass stays visible. Inverse-variance weights make the estimator worse by a factor of two to six under the heteroskedastic design and do not help on the listed chains. The weights move the fit toward the cheap wing options, whose premiums are precise, and away from the body of the distribution, where the density lives; the penalty rule, which was calibrated for unit weights, is also disturbed. We take this as evidence that the unweighted spline with a global penalty is the right default, not that weighting could never help, and leave a re-calibrated weighted version for future work.

5.6. The competitors at our bandwidth

If the advantage of the procedure lay in the bandwidth rule rather than in the estimator, handing the rule to the competitors should close the gap. Table 3 contains the rows "at (11)", in which each competitor is evaluated at our rule bandwidth, converted to its own unit (divided by F for the moneyness kernel; multiplied by $\sqrt{7}$ for the quartic half-width, which equates the standard deviations), and Table 9 repeats the exercise on the listed chains. The transfer fails in both directions. At our bandwidth the competitors are two to sixteen times worse than our estimator in the simulations and are as far from their own oracles as under their published rules; on the listed chains their hold-out is between 1.7 and seven times ours, and for the call-space methods orders of magnitude more, and most of their densities acquire extra modes. The reason is visible in the oracle bandwidths of Table 5 and in Table 10: the implied-volatility smoothers want bandwidths of a few strike gaps, a quarter to a third of ours, because they smooth an almost flat function and pay for every bit of width in bias, whereas our estimator smooths the integrated-tail distribution, whose second derivative is the target, with a sixth-order kernel whose bias is small at a width of one to two percent of the forward. The rule (11) is the right scale for that construction and the wrong scale for theirs. The two pieces, the kernel and the rule, are therefore not separable: the kernel without the rule has no feasible tuning that reaches its oracle, as Section 5.5 shows for GCV and Table 3 for cross-validation, and the rule without the kernel is of no use to the other estimators. That is what we mean by a procedure.

Table 9: Listed chains, each competitor at the rule bandwidth (11) of our estimator. Columns as in Table 11.

		Hold-out			In-sample				
Chain	Method	pts	hs	bp	hs	Mass	Neg.	Modes	TV
SPX Dec 2026	PCA, at (11)	**0.059**	0.39	89	0.34	1.00	-0.000	1	1.02
	AS–Lo, at (11)	0.598	1.04	**12**	0.92	1.01	0.015	4	1.70
	AS–Duarte, at (11)	1.704	2.44	15	2.10	1.00	0.000	4	1.04
	GHS IV, at (11)	0.068	**0.39**	23	0.07	1.01	0.000	1	1.04
	GHS call, at (11)	14.250	109.56	1116	0.07	1.00	0.000	1	1.02
SPX Mar 2027	PCA, at (11)	**0.081**	**0.46**	101	0.36	1.00	-0.000	2	1.01
	AS–Lo, at (11)	0.486	0.81	23	0.71	1.01	0.007	1	1.00
	AS–Duarte, at (11)	2.649	3.25	**17**	2.80	1.00	-0.000	2	1.00
	GHS IV, at (11)	0.185	1.07	55	0.06	1.01	-0.000	1	1.04
	GHS call, at (11)	30.763	203.95	1609	0.07	1.00	-0.000	1	1.02
NDX Dec 2026	PCA, at (11)	1.258	0.15	166	0.13	1.00	-0.000	3	1.20
	AS–Lo, at (11)	6.965	0.46	47	0.43	1.02	0.019	3	2.10
	AS–Duarte, at (11)	10.013	0.62	**22**	0.56	1.00	0.000	5	1.27
	GHS IV, at (11)	**1.235**	**0.13**	57	0.09	1.02	0.000	3	1.08
	GHS call, at (11)	41.438	8.89	884	0.07	0.98	0.000	3	1.07
RUT Dec 2026	PCA, at (11)	**0.117**	**0.46**	148	0.36	1.00	-0.000	2	1.01
	AS–Lo, at (11)	0.386	0.82	**21**	0.74	1.01	0.007	2	1.55
	AS–Duarte, at (11)	1.292	2.20	28	1.91	1.00	0.000	5	1.18
	GHS IV, at (11)	0.561	1.79	75	0.04	1.01	0.000	1	1.02
	GHS call, at (11)	23.602	101.48	2120	0.04	1.00	0.000	1	1.02

6. Listed Chains

6.1. Data

The listed comparison uses European, AM-settled index options from Cboe's delayed-quotes feed (`cdn.cboe.com`). Each file is dated 23 September 2026; the S&P 500 (SPX) snapshot is time-stamped 03:56 Eastern, the Nasdaq-100 (NDX) and Russell 2000 (RUT) snapshots 03:44 Eastern. At those hours the quotes are the overnight or previous-session book for the contracts that trade in extended hours and the previous close for the others, and the cash index level in the file is the previous close; we use the snapshot as a static chain and make no claim about intraday pricing. Only the standard monthly roots SPX, NDX, and RUT are kept, not the weekly roots. Time to expiry is $T = d/365.25$ with d the calendar-day count from the snapshot date, and the discount rate is $r = 0.04$ at every maturity; the forward is backed out from put–call parity, so the rate affects only the discounting of the premiums and enters every method identically.

A quote is retained if the bid is positive, the ask is at least the bid, the mid is at least 0.25, and the spread is at most max(15, $0.002S_0$, $0.5 \times$ mid). Among the remaining put–call pairs, the forward is the median of $K + e^{rT}(C - P)$ over strikes within 50 points of the pair closest to the cash index on SPX and within 0.65% of spot on NDX and RUT, and the dividend yield is $q = r - \log(F/S_0)/T$. The working slice is out of the money: the put mid for $K \le F$ and the call mid for $K \ge F$, the other side filled by parity, and puts with $K < 0.7F$ and mid below 0.50 dropped. The bid–ask spread of the quoted side is kept with each strike. Table 10 records the chains: the December 2026 and March 2027 SPX quarterlies and the December 2026 NDX and RUT quarterlies, with 99 to 444 strikes, maturities of 0.24 and 0.49 years, and median half-spreads of half a point on SPX and RUT and fourteen points on NDX.

Table 10: The four listed chains and the tuned bandwidths. n is the number of strikes after the screens, $\hat{\sigma}$ the at-the-money implied volatility in percent, hs the median half-spread in index points. The bandwidths are in strike units for every method, including the moneyness bandwidth of Aït-Sahalia and Lo multiplied by F; for the local cubics of Grith et al. they are the half-width of the quartic window. Ours, rule is the widened bandwidth (11); c is the penalty (12).

						Bandwidth							
Chain	n	F	T	$\hat{\sigma}$	hs	Ours, rule	Ours, CV	PCA	AS–Lo	AS–D	GHS IV	GHS call	c
SPX Dec 2026	444	7836	0.235	13.4	0.50	80.0	20.4	225.0	20.4	20.4	363.5	587.3	4.25e-6
SPX Mar 2027	221	7918	0.485	14.4	0.45	133.6	40.4	350.2	40.4	40.4	719.2	1162.0	0.00e0
NDX Dec 2026	312	31025	0.235	20.1	13.70	503.7	153.5	647.3	153.5	153.5	647.3	3469.7	2.14e-5
RUT Dec 2026	99	2912	0.235	18.3	0.50	47.8	43.6	144.8	16.7	16.7	297.4	297.4	1.05e-5

Every method is fitted to the same input: the quoted calls replaced by the nearest vector in Euclidean distance that is decreasing and convex in strike, with slopes in $[-e^{-rT}, 0]$ and each coordinate at least the intrinsic value. This is the unpenalized projection of Yatchew and Härdle (2006) and the first step of Aït-Sahalia and Duarte (2003). Puts of the projected call follow by parity.

6.2. Methods and tuning protocol

Five estimators from the literature are compared with ours. *PCA* is the positive convolution approximation of Bondarenko: a Gaussian kernel on the spot convolved with a discrete probability on equally spaced nodes from the first quoted strike to the last, with the node spacing equal to the bandwidth and the count between 12 and 61; the weights are nonnegative, sum to one, have mean F, and minimize the squared pricing error, each node pricing a call by the Bachelier formula.

AS–Lo is the estimator of Aït-Sahalia and Lo on one maturity, where the semiparametric regression reduces to Nadaraya–Watson of Black–Scholes implied volatility on moneyness K/F with a Gaussian kernel; prices are Black–Scholes at the smoothed volatility and the density is the strike derivative of that call. *AS–Duarte* is the shape-constrained estimator of Aït-Sahalia and Duarte: the projection above followed by a Gaussian local linear regression; the price is the local level and the density is e^{rT} times the strike derivative of the local slope, scaled to unit mass and shifted to mean F as in their Section 3.6. *GHS call* and *GHS IV* are the local cubics of Grith et al. with the quartic kernel, of the call divided by the spot and of implied volatility against strike; the density is e^{rT} times the second derivative of the local cubic for the call, and the implied-volatility curvature formula for the smile.

Each of these has one bandwidth. The common protocol tunes it by even/odd cross-validation of the out-of-the-money pricing error: the strikes are split into even and odd positions, the method is fitted on one half and scored on the other, the two directions are averaged, and the bandwidth minimizing that average on a geometric grid of 19 points from 0.04 to 3 times $F\hat{\sigma}_{\mathrm{ATM}}\sqrt{T}$, floored at one and a half median strike gaps, is kept. The same grid, in strike units, is used for every method; for AS–Lo it is divided by F. Our estimator is tuned the same way ("Ours, CV"), with the penalty (12) applied at each candidate bandwidth, and is also reported at the rule (10)–(11) ("Ours, rule"). The source papers' own choices are reported as well: the Appendix A bandwidth of Aït-Sahalia and Lo, which on a single maturity is about one standard deviation of moneyness; the Fan and Gijbels (1996) plug-in of Aït-Sahalia and Duarte, Section 3.5, which uses the level rate $n^{-1/5}$; and the leave-one-out bandwidth of Grith et al. together with twice that width, which is what their figures use. Those rows are in Appendix C.

Hold-out is nested. For each of the two even/odd directions, the projection and the tuning are recomputed on the training half alone, including the rule bandwidth (11) with the training n and the training roughness; the fitted call is then scored on the held-out half against the quoted mids. Errors are reported in three units: index points, the quoted half-spread of the held-out option, and Black–Scholes implied volatility in basis points, each as a root mean square over the held-out strikes. The half-spread unit makes chains of different levels comparable and sets the error against the market's own precision; an error of one means the fitted price sits at the quoted bid or ask. The implied-volatility unit is the one practitioners use, and a fitted premium so far off that no implied volatility exists is counted as one hundred percent. Densities are evaluated on $[\max(50, 0.2F), 2.4F]$, and we report their positive mass, their negative mass $\int \hat{f}_{\mathbb{Q}}^{-}$, the number of modes on $[0.60F, 1.30F]$

(local maxima above a fifth of the global one), and the total-variation score (8) on that window.

6.3. Results

Table 11 reports the four chains. Three observations organize it. First, under the common tuning the rule estimator has the smallest nested hold-out error of the competing methods on every chain in half-spread units, and it is within four thousandths of a half-spread of its own cross-validated version, which is ahead by that margin on three chains; in index points the two versions of our estimator are likewise within a few thousandths of a point of each other and ahead of the rest; in implied volatility the rule is the smallest or tied on every chain. The hold-out is 0.07–0.11 half-spreads: the fitted price of a held-out option sits about a tenth of the way from the mid to the quoted bid or ask. The nearest competitor is the cross-validated implied-volatility cubic of Grith et al., at 0.10–0.38 half-spreads, followed by PCA at 0.14–0.34; the kernel smoothers of Aït-Sahalia and Lo and Aït-Sahalia and Duarte, and the call cubic, are further off. Second, the price cross-validation that tunes the competitors selects the floor of the grid for the implied-volatility kernel and the local linear on every chain, and their densities are not usable: the implied-volatility kernel has nine to thirty modes and negative mass of up to 1.1 on every chain, the local linear eleven and twenty modes on two of them, and even the implied-volatility cubic has eleven modes on NDX. Pricing error between neighbouring strikes does not penalize a rough second derivative, which is why Section 3.5 sets the bandwidth by a rate rather than by cross-validation; the same effect shows in "Ours, CV", which prices as well as the rule but has four modes on NDX and two on SPX March. Third, among the methods whose densities are well behaved, the differences in pricing are small in economic terms: every one of them prices inside the quoted spread in sample, and the hold-out of the best three differs by a tenth of a half-spread. The case for the estimator on listed data is therefore that it delivers one mode, unit mass and no negative mass to three decimals on every chain, with the smallest pricing error, rather than that it prices dramatically better.

Table 10 lists the tuned bandwidths. The rule bandwidth (11) is one to two percent of the forward; the roughness (9) at the base bandwidth is zero on SPX March, 0.04 on the two other SPX and RUT chains and 0.21 on NDX, so the widening is at most three percent, and the listed chains are, for the rule, close to the exact-quote regime. The cross-validated version is narrower on every chain, by a factor of three to four on SPX December and NDX, which is the under-smoothing just described. The penalty (12) is zero on SPX March and between

Table 11: Listed chains, common cross-validation. Hold-out is the nested even/odd error of the fitted out-of-the-money premium against the quoted mid, in index points, in units of the quoted half-spread (hs), and in implied volatility (bp). In-sample is the full-chain error in half-spreads. Mass is $\int \max(\hat{f}_{\mathbb{Q}}, 0)$ and Neg. is $\int \max(-\hat{f}_{\mathbb{Q}}, 0)$ on $[\max(50, 0.2F), 2.4F]$; Modes and TV are on $[0.60F, 1.30F]$. The smallest hold-out in each unit is in bold.

		Hold-out			In-sample				
Chain	Method	pts	hs	bp	hs	Mass	Neg.	Modes	TV
SPX Dec 2026	Ours, rule	0.038	0.11	6	0.07	1.00	0.000	1	1.00
	Ours, CV	**0.036**	**0.10**	**6**	0.07	1.00	0.000	1	1.02
	PCA, CV	0.054	0.34	54	0.29	1.00	-0.000	1	1.00
	AS–Lo, CV	0.093	0.22	9	0.13	1.52	0.557	30	10.78
	AS–Duarte, CV	0.261	0.43	338	0.15	1.01	0.015	1	1.02
	GHS IV, CV	0.059	0.19	13	0.07	1.00	0.000	1	1.01
	GHS call, CV	1.415	2.17	15	0.49	1.00	0.001	1	1.00
SPX Mar 2027	Ours, rule	0.051	0.08	**3**	0.07	1.00	0.000	1	1.00
	Ours, CV	**0.044**	**0.08**	3	0.06	1.00	0.000	2	1.02
	PCA, CV	0.075	0.33	46	0.25	1.00	-0.000	1	1.00
	AS–Lo, CV	0.176	0.38	20	0.11	1.37	0.369	9	4.98
	AS–Duarte, CV	0.900	1.14	10	0.27	1.00	0.000	11	4.38
	GHS IV, CV	0.137	0.38	35	0.10	1.00	-0.000	1	1.00
	GHS call, CV	2.246	3.07	22	1.42	1.00	0.000	1	1.00
NDX Dec 2026	Ours, rule	1.085	0.08	**10**	0.07	1.00	0.000	1	1.00
	Ours, CV	**1.082**	**0.08**	10	0.07	1.00	0.000	4	1.06
	PCA, CV	1.298	0.14	101	0.13	1.00	-0.000	2	1.03
	AS–Lo, CV	3.165	0.22	39	0.11	1.59	0.591	19	7.80
	AS–Duarte, CV	4.090	0.27	407	0.08	1.04	0.037	1	1.00
	GHS IV, CV	1.220	0.10	17	0.09	1.02	0.000	11	2.13
	GHS call, CV	2.038	0.15	11	0.11	1.01	0.000	1	1.00
RUT Dec 2026	Ours, rule	**0.029**	**0.07**	5	0.05	1.00	0.001	1	1.00
	Ours, CV	0.030	0.07	5	0.05	1.00	0.001	1	1.00
	PCA, CV	0.065	0.29	66	0.27	1.00	-0.000	1	1.00
	AS–Lo, CV	0.245	0.51	17	0.15	2.12	1.119	13	7.75
	AS–Duarte, CV	0.781	1.36	19	0.25	1.00	0.000	20	7.81
	GHS IV, CV	0.088	0.16	**5**	0.06	1.03	0.000	1	1.00
	GHS call, CV	1.285	2.41	66	0.25	1.01	0.001	1	1.00

4×10^{-6} and 2×10^{-5} on the other three chains. Figure 3 draws the December chains. The rule estimator, PCA and the doubled implied-volatility cubic agree on the body of the density and differ in the left tail, where the quotes are sparse; the plug-in local linear is visibly oversmoothed.

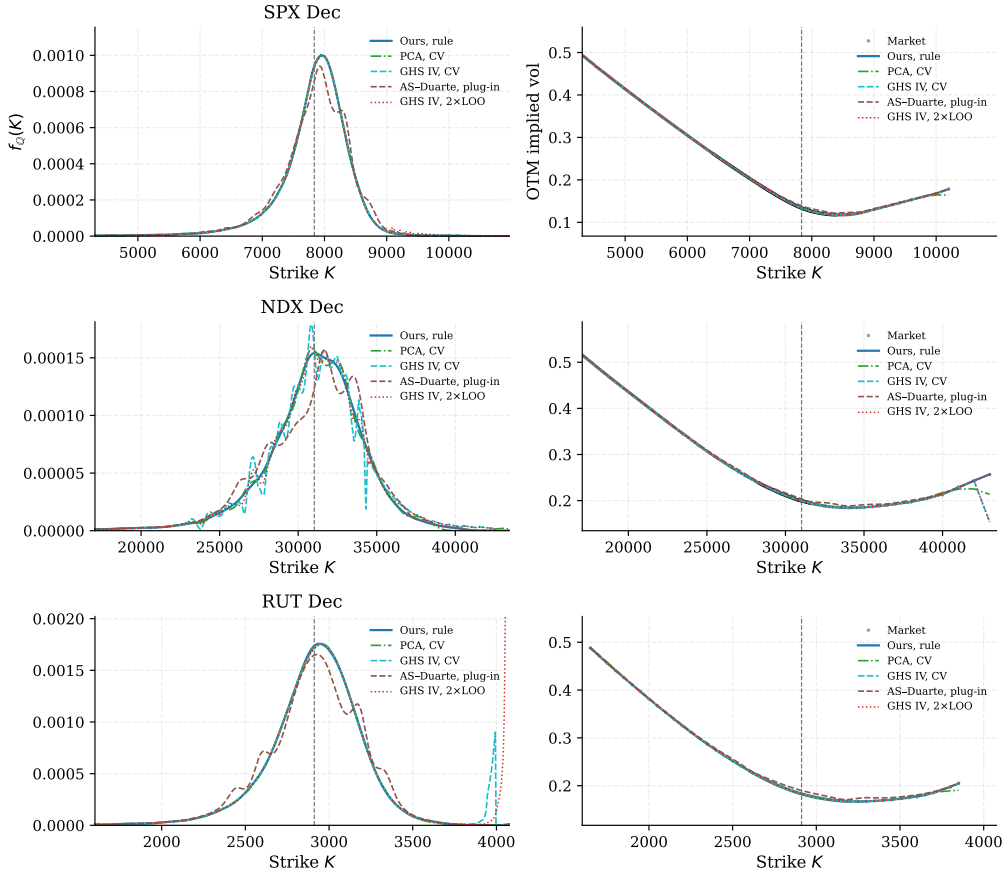


Figure 3: SPX, NDX, and RUT, 18 December 2026, snapshot 23 September 2026. Each curve is fitted to the projected call of Section 6.1. Left: $\max(\hat{f}_{\mathbb{Q}}, 0)$ for the rule estimator, PCA, the implied-volatility cubic of Grith et al. at the cross-validated width, the local linear of Aït-Sahalia and Duarte at the plug-in, and the implied-volatility cubic at twice the leave-one-out width, as in Grith et al.. Right: out-of-the-money implied volatility of the market and of the same fits. The dashed line is the forward.

6.4. Further chains

The constants of Section 3.5 were settled on the four chains of Table 11 and on the exact designs. As a check on that choice, every remaining standard monthly

expiry in the four snapshots, SPX, NDX and RUT of 23 September 2026 and the Dow Jones (DJX) of 19 September 2026, with at least forty strikes after the screens and maturities between two weeks and two years, is scored with the rules held fixed: 31 chains. Table 12 summarizes the nested hold-out in half-spread units for the cross-validated methods; the full table is in Appendix D. These quotes are not in the replication files, because under Cboe's data policies a recipient of delayed quotes may not redistribute them.

Table 12: Summary over the 31 further chains of Appendix D. Hold-out is the nested even/odd error in half-spread units; rank is the mean rank of the method across chains (one is best); wins is the number of chains on which the method has the smallest hold-out; bp is the median hold-out in implied volatility; unimodal is the share of chains on which the fitted density has one mode; neg. is the mean negative mass.

	Hold-out, hs						
Method	mean	median	rank	wins	bp	unimodal, %	neg.
Ours, rule	0.17	0.10	2.35	5	15	90	0.001
Ours, CV	0.15	0.11	1.97	16	8	39	0.001
PCA, CV	0.45	0.32	4.94	0	51	87	0.000
AS–Lo, CV	0.34	0.33	4.58	0	30	3	0.467
AS–Duarte, CV	2.85	0.56	5.94	0	30	16	7128.970
GHS IV, CV	0.21	0.12	2.45	10	9	81	0.000
GHS call, CV	1.79	0.79	5.77	0	25	94	0.008

Over the 31 further chains the ranking of Table 11 is preserved in the average but not on every chain. The rule estimator has the smallest mean and median hold-out (0.17 and 0.10 half-spreads) and the best mean rank (2.35) among the rule-based and cross-validated methods other than its own cross-validated version, which has a slightly better mean rank but is unimodal on only 39% of the chains. The implied-volatility cubic is close behind (0.21 and 0.12, rank 2.45) and is the better of the two on the long-dated NDX chains, where the quoted mesh widens to a thousand points and the rule bandwidth is narrow relative to the gaps; PCA is third. The rule estimator is unimodal on 90% of the chains; the exceptions are three chains of a year or more to expiry on which the fitted density has a second mode of about half the height of the first, which the rule, by design, does not remove. Its negative mass is 0.001 on average. The implied-volatility kernel and the local linear, tuned by price cross-validation, are unimodal on 3% and 16% of chains.

7. Conclusion

The normalized call of a single maturity is the survival function of the integrated-tail distribution of the underlying, and the risk-neutral density is minus the forward times the second derivative of that distribution function. The estimator of this paper fits that function by a thriced cubic smoothing spline, completes the unquoted wings, and smooths the derivative of the spline by a sixth-order Gaussian jackknife kernel; the convolution is in closed form, the density and the fitted call are two readings of one object, and the bandwidth and penalty follow from one number read off the chain, the roughness of the thriced interpolant, with no search. The estimator is in-fill consistent under additive noise.

The evidence is as follows. On exact quotes the wings and the higher-order kernel reduce the error of a plain spline by several orders of magnitude. On noisy quotes the estimator is the most accurate, or statistically tied with the most accurate, in every Heston and Bates design at the central noise level and in all but one at the other two. Variance gamma is its weak target: the density of S_T under a pure-jump process with a quarter year to expiry has a cusp at the forward, which the sixth-order kernel's negative sidelobes resolve less well than a positive mixture or a smoother of the nearly flat implied-volatility smile does, and there the positive convolution approximation and the implied-volatility kernel of Aït-Sahalia and Lo are better by factors of 1.4 to 2.3 on the regular meshes. On listed index chains it has the smallest nested out-of-sample pricing error of the competing methods on each chain in index points, half-spreads and implied volatility, and it is the only method among those compared that delivers one mode, unit mass, and no negative mass on every chain; the same ordering holds on average over 31 further chains scored with the rules fixed. Two further findings qualify the claim. At infeasible oracle bandwidths the implied-volatility smoothers are more accurate than our kernel on regular meshes, and pricing cross-validation under-smooths every density estimator on dense chains, ours included; what distinguishes the procedure proposed here is that its rule-based tuning reaches within a factor of two of its oracle, while no feasible tuning does so for the competitors, and the rule does not transfer to them. The contribution is the whole: the transform, the thriced spline, the higher-order kernel, and the roughness that sets both constants.

Several extensions are natural. The theorem gives consistency; a rate and a limiting distribution for the penalized estimator, and hence confidence bands, would make the estimator usable for inference about tail probabilities and pricing kernels. The tail completion is linear and could be replaced by a parametric tail fitted to the last quotes, as in Figlewski (2010). Finally, a panel of dates rather than

one snapshot would allow the stability of the estimate over time to be assessed, which is the criterion most of the applied literature uses.

Appendix A. Proof of Theorem 1

Throughout, C denotes a constant that does not depend on m and may change from line to line, $[A', B']$ is a fixed compact subset of (A, B), $\varepsilon = \min(A' - A, B - B') > 0$, and $\|\cdot\|$ is the supremum norm on $[A', B']$. All limits are as $m \to \infty$. We write $L = L_1$ for the thriced kernel at unit bandwidth, so that $L_h(u) = h^{-1}L(u/h)$, and note that L is even, integrates to one, has vanishing moments of orders two and four, and that L, L', L'' and L''' are bounded by multiples of $e^{-u^2/8}$.

Reduction. The estimator is $\hat{f}^{\mathrm{th}}_{\mathbb{Q}} = -F\,(T'_\lambda * L_h)' = -F\,(T_\lambda * L''_h)$, where $T_\lambda = I - (I - S_\lambda)^3$ is the thriced spline (7) built from the natural cubic smoothing spline S_λ through the knot values (K_i, P_i), the two tail knots included, with $\lambda = cF^3$ and c given by (12); the second form follows by moving both derivatives onto the kernel. Both S_λ and T_λ are linear in the data, $T_\lambda = T_\lambda P + T_\lambda \eta$, where $T_\lambda P$ is the thriced spline through the exact values and $T_\lambda \eta$ that through the noise. It therefore suffices to show that

$$\left\|(T_\lambda P) * L''_h + P''\right\| \;\to\; 0 \quad \text{and} \quad \left\|(T_\lambda \eta) * L''_h\right\| \;\to_p\; 0, \tag{A.1}$$

where λ is random in the second statement and h is random in both. Steps 1 to 5 take h deterministic with $h \to 0$, $\bar{\delta}/h \to 0$ and $nh^5/\log n \to \infty$; Step 6 removes that restriction. At $\lambda = 0$, $T_0 = S_0$ is the interpolant. We first treat $\lambda = 0$, then show that the random λ is small, and then extend both statements to the random penalty.

Preliminaries on splines. Four facts about natural cubic splines on a quasi-uniform mesh are used; all are classical. (S1) If $g \in C^4[K_1, K_m]$, the natural interpolating spline $S_0 g$ satisfies $\|(S_0 g)^{(k)} - g^{(k)}\|_{\infty,[A',B']} \le C\bar{\delta}^{4-k}$ for $k = 0, 1, 2$: the $O(\bar{\delta}^{4-k})$ interior bound is the Hall–Meyer bound for complete splines (Hall and Meyer, 1976), and the perturbation caused by the natural end conditions decays geometrically away from the end knots, at the rate $2 - \sqrt{3}$ per knot (de Boor, 2001, Ch. IV), so it is exponentially small on $[A', B']$. (S2) The interpolating spline is $S_0 g = \sum_i g(K_i)\phi_i$, where the cardinal functions satisfy $|\phi_i^{(k)}(x)| \le C\bar{\delta}^{-k}\, r^{|x-K_i|/\bar{\delta}}$ for $k = 0, 1, 2$ and some $r < 1$ (Demko, 1977). (S3) The smoothing spline is $S_\lambda g = \sum_i g(K_i)\phi_i^\lambda$, and the weight functions ϕ_i^λ are localized at the larger of the two scales $\bar{\delta}$ and $\beta = (\lambda\delta)^{1/4}$: $|\phi_i^\lambda(x)| \le C\,\frac{\bar{\delta}}{\bar{\delta}\vee\beta}\exp(-\gamma|x - K_i|/(\bar{\delta}\vee\beta))$ with $\gamma > 0$

and C not depending on λ (Silverman, 1984; Messer and Goldstein, 1993). (S4) For $g \in C^4$ and λ in a bounded set, $\|(S_\lambda g)^{(k)} - g^{(k)}\|_{\infty,[A',B']} \le C(\beta^{4-k} + \bar\delta^{4-k})$ for $k = 0, 1, 2$ (Ragozin, 1983).

Two facts about the kernel are used. (K1) If $g \in C^6$ on a neighbourhood of $[A', B']$, then $\|g * L_h'' - g''\| \le Ch^4\|g^{(6)}\|_\infty$, and if $g \in C^8$ then $\|g * L_h'' - g''\| \le Ch^6\|g^{(8)}\|_\infty$; both follow from $g * L_h'' = g'' * L_h$, a Taylor expansion of g'' to order four or six, and the vanishing moments of L up to order five. (K2) For $k \le 3$, $\int |L_h^{(k)}| \le Ch^{-k}$ and $|L_h^{(k)}(u)| \le Ch^{-1-k}e^{-u^2/8h^2}$.

Step 1: the deterministic part at penalty zero. By (S1) with $g = P$ and (K2), $\|(S_0P - P) * L_h''\| \le \|S_0P - P\|_{\infty,[A'-\varepsilon/2,B'+\varepsilon/2]} \int |L_h''| + Ch^{-3}e^{-\varepsilon^2/32h^2}\|P\|_\infty \le C\bar\delta^4h^{-2} + o(1)$, where the second term collects the part of the convolution that reaches outside $[A' - \varepsilon/2, B' + \varepsilon/2]$, including the two tail knots. By (K1) with $g = P \in C^6$, $\|P * L_h'' - P''\| \le Ch^4$. Hence $\|(S_0P) * L_h'' - P''\| \le C(\bar\delta^4h^{-2} + h^4) + o(1) \to 0$, using $\bar\delta/h \to 0$. When $P \in C^8$, that is when $f_{\mathbb{Q}} \in C^6$, the bias is $O(\bar\delta^4h^{-2} + h^6)$, which is the statement of Remark 1.

Step 2: the noise at penalty zero. Write $Z_0(x) = ((S_0\eta) * L_h'')(x) = \sum_i \eta_i\, w_i(x)$ with $w_i = \phi_i * L_h''$. By (S2) and (K2),

$$|w_i(x)| \;\le\; \int |\phi_i(y)|\,|L_h''(x-y)|\,\mathrm{d}y \;\le\; C\bar\delta\, h^{-3}\,\Lambda((x-K_i)/h),$$

with $\Lambda(u) = \sup_{|v-u|\le 1} e^{-v^2/8} + e^{-u^2/16}$, because $\int|\phi_i| \le C\bar\delta$ and ϕ_i is concentrated within $O(\bar\delta\log(h/\bar\delta)) \ll h$ of K_i. Since Λ is bounded and integrable, quasi-uniformity gives

$$\sum_i w_i(x)^2 \;\le\; C\bar\delta^2h^{-6}\sum_i \Lambda((x-K_i)/h)^2 \;\le\; C\bar\delta\,h^{-5} \;\le\; \frac{C}{nh^5}, \tag{A.2}$$

uniformly in $x \in [A', B']$. The same computation with L_h''' in place of L_h'' bounds the weights of Z_0' and gives $\sum_i w_i'(x)^2 \le C/(nh^7)$. By Assumption 1(iii), $Z_0(x)$ is sub-Gaussian with proxy $\tau^2\sum_i w_i(x)^2$, and so is every increment $Z_0(x) - Z_0(x') = \sum_i \eta_i(w_i(x) - w_i(x'))$, with proxy $\tau^2\sum_i(w_i(x) - w_i(x'))^2 \le \tau^2|x-x'|^2\sup_y\sum_i w_i'(y)^2 \le C\tau^2|x-x'|^2/(nh^7)$. Let $x_1, \dots, x_N$ be a net of $[A', B']$ with spacing h, so $N \le C/h$. The maximal inequality for sub-Gaussian variables gives $\mathbb{E}\max_j|Z_0(x_j)| \le C\tau\sqrt{\log N/(nh^5)}$. On each cell $[x_j, x_{j+1}]$ the process has increments with proxy at most $C\tau^2|x-x'|^2/(nh^7)$, so its diameter in the sub-Gaussian metric is $C\tau h/(nh^7)^{1/2} = C\tau/(nh^5)^{1/2}$ and its covering number at scale u is at most $C\tau/(u(nh^5)^{1/2})$; Dudley's entropy bound (van der Vaart and Wellner, 1996, Corollary 2.2.8) gives

$\mathbb{E}\sup_{x\in[x_j,x_{j+1}]}|Z_0(x)-Z_0(x_j)| \le C\tau/(nh^5)^{1/2}$, and the same corollary bounds the maximum of the N cell oscillations by $\sqrt{\log N}$ times that quantity. Altogether

$$\mathbb{E}\|Z_0\| \;\le\; C\tau\sqrt{\frac{\log n}{nh^5}} \;\to\; 0. \tag{A.3}$$

With Step 1 this proves the theorem at penalty zero, which is the case of exact quotes and the case $c = 0$.

Step 3: the random penalty is small. The penalty (12) is $c = x(\rho)\,h^4/(\delta F^3)$ with $\rho = (\mathrm{TV}-\mathrm{TV}_*)_+/(v_*-1)$, where TV is the score (8) of the positive part of the thriced interpolant $\hat f_0 = -F\,(S_0 * L_h'')$ on the window $W = [0.60F, 1.30F]$, and $x(\rho) = t/(1-t)$, $t = (\rho/(1+\rho))^{1/3}$, satisfies $x(\rho) \le 1+3\rho$ for all $\rho \ge 0$, so a bound on ρ is a bound on x. Write $\hat f_0 = f_0 + \zeta$ with $f_0 = -F(S_0P) * L_h''$ deterministic and $\zeta = -FZ_0$. The completed variation of the positive part of a function on W is at most the completed variation of the function itself, $|g(a)| + \int_W |g'| + |g(b)|$, and the maximum of $\hat f_0$ on W is, by Steps 1 and 2, within $o_p(1)$ of $\max_W f_{\mathbb{Q}} > 0$. Hence

$$\rho \;\le\; \frac{1}{v_*-1}\,\frac{2\|\hat f_0\|_{\infty,W} + \int_W |f_0'| + \int_W |\zeta'|}{2\max_W \hat f_0} \;=\; O_p\Big(1 + \int_W |\zeta'|\Big),$$

and $\mathbb{E}\int_W|\zeta'| \le |W|\sup_x(\mathbb{E}\zeta'(x)^2)^{1/2} \le C\tau(nh^7)^{-1/2}$ by the bound on $\sum_i w_i'(x)^2$, since f_0' is bounded by Step 1. Therefore $\rho = O_p(1+(nh^7)^{-1/2})$, hence $x(\rho) = O_p(1+(nh^7)^{-1/2})$, and, since $\delta \asymp 1/n$,

$$\beta^4 \;=\; cF^3\delta \;=\; x(\rho)\,h^4 \;=\; O_p(h^4 + h^4(nh^7)^{-1/2}) \;=\; O_p(h^4 + (h/n)^{1/2}), \tag{A.4}$$

so $\beta = o_p(1)$, because $h/n = h^6/(nh^5) \to 0$. In particular there is a deterministic sequence $b_m = ((h^4 + (h/n)^{1/2})\log n)^{1/4} \to 0$ with $\mathbb{P}(\beta > b_m) \to 0$. Since $\rho \ge 0$, on the event $E_m = \{\beta \le b_m\}$ the penalty lies in $[0, \lambda_m]$ with $\lambda_m = b_m^4/\delta$.

Step 4: the deterministic part at the random penalty. On E_m, (S4) with $g = P$ and $\lambda \le \lambda_m$ gives $\|S_\lambda P - P\|_{\infty,[A'-\varepsilon/2,B'+\varepsilon/2]} \le C(b_m^4 + \bar\delta^4)$. The thriced spline inherits the bound: $P - T_\lambda P = (I-S_\lambda)^3P = (I-S_\lambda)^2(P-S_\lambda P)$, and by (S3) the operator $I-S_\lambda$ is bounded on the supremum norm uniformly in λ and m, because $\sum_i|\phi_i^\lambda(x)| \le C$ by the localization and quasi-uniformity; hence $\|T_\lambda P - P\|_{\infty,[A'-\varepsilon/2,B'+\varepsilon/2]} \le C(b_m^4+\bar\delta^4)$ as well. The argument of Step 1, with $T_\lambda P$ in place of S_0P, then yields

$$\big\|(T_\lambda P) * L_h'' + P''\big\| \;\le\; C(b_m^4h^{-2} + \bar\delta^4h^{-2} + h^4) + o(1) \;\to\; 0,$$

because $b_m^4h^{-2} = (h^2 + (nh^5)^{-1/2}h^{1/2})\log n \to 0$ under $nh^5/\log n \to \infty$. The bound is uniform in $\lambda \in [0,\lambda_m]$, so it holds at the random penalty on E_m, and $\mathbb{P}(E_m^c) \to 0$.

Step 5: the noise at the random penalty. Let $Z_\lambda(x) = ((T_\lambda\eta) * L_h'')(x)$. We bound $\sup\{|Z_\lambda(x)| : x \in [A', B'],\ 0 \le \lambda \le \lambda_m\}$, which dominates the noise at the random penalty on E_m. Since $b_m^4/h^4 = (1 + (nh^7)^{-1/2}) \log n$ need not tend to zero, the localization scale of ϕ_i^λ in (S3) can exceed h, and we argue through the operator form instead. On the knot values the smoothing spline acts as $(I + \lambda\Omega)^{-1}$, with Ω the symmetric nonnegative roughness matrix (Green and Silverman, 1994); let $\Omega = \sum_j \nu_j u_j u_j^\top$ be its eigendecomposition (the Demmler–Reinsch basis), with $\nu_j \ge 0$. The thriced spline acts as $p((I + \lambda\Omega)^{-1})$ with $p(s) = 1 - (1 - s)^3$, a polynomial that maps $[0, 1]$ onto $[0, 1]$ with $p'(s) = 3(1 - s)^2 \le 3$. The map from knot values to $Z_\lambda(x)$ is linear, $Z_\lambda(x) = \langle \eta, p((I + \lambda\Omega)^{-1})a_x \rangle$, where a_x is the vector with entries $w_i(x) = (\phi_i * L_h'')(x)$ of Step 2, so the sub-Gaussian proxy of $Z_\lambda(x)$ is $\tau^2 \| p((I + \lambda\Omega)^{-1})a_x \|^2 \le \tau^2 \|a_x\|^2 \le C\tau^2/(nh^5)$ by (A.2): at any fixed penalty the variance is at most the penalty-zero variance, uniformly in x. The same holds for $\partial_x Z_\lambda$, with $C\tau^2/(nh^7)$. For increments in the penalty, $\partial_{\log\lambda}(I + \lambda\Omega)^{-1} = -\lambda\Omega(I + \lambda\Omega)^{-2}$, whose eigenvalues $-\lambda\nu_j/(1 + \lambda\nu_j)^2$ lie in $[-1/4, 0]$, so those of $\partial_{\log\lambda}\, p((I + \lambda\Omega)^{-1})$ lie in $[-3/4, 0]$; the mean value theorem applied coordinatewise in the eigenbasis shows that the proxy of $Z_\lambda(x) - Z_{\lambda'}(x)$ is at most $\tau^2 |\log\lambda - \log\lambda'|^2 \|a_x\|^2$ for $\lambda, \lambda' > 0$. Hence the process $(x, \ell) \mapsto Z_{e^\ell}(x)$ on $[A', B'] \times [\log\lambda_{\min}, \log\lambda_m]$ is sub-Gaussian with respect to the metric

$$d((x, \ell), (x', \ell')) \;=\; \frac{C\tau}{(nh^5)^{1/2}} \Big(\frac{|x - x'|}{h} + |\ell - \ell'| \Big).$$

Take $\lambda_{\min} = \beta_{\min}^4/\delta$ with $\beta_{\min} = h/\log n$; then $\log(\lambda_m/\lambda_{\min}) = 4\log(b_m \log n/h) = O(\log n)$, because b_m/h is bounded by a power of n by (A.4). Cover the rectangle by cells of side h in x and one in ℓ; there are $N \le C(\log n)/h$ cells, each of d-diameter at most $C\tau(nh^5)^{-1/2}$, and the entropy integral over a cell is bounded by a constant times its diameter. The maximal inequality over the cell centres and Dudley's bound within cells give, exactly as in Step 2,

$$\mathbb{E} \sup\{|Z_\lambda(x)| : x \in [A', B'],\ \lambda_{\min} \le \lambda \le \lambda_m\} \;\le\; C\tau \sqrt{\frac{\log N}{nh^5}} \;\le\; C\tau \sqrt{\frac{\log n}{nh^5}} \;\to\; 0.$$

It remains to cover $0 \le \lambda \le \lambda_{\min}$. There $p((I + \lambda\Omega)^{-1}) - I = -(\lambda\Omega(I + \lambda\Omega)^{-1})^3$ has eigenvalues $-(\lambda\nu_j/(1 + \lambda\nu_j))^3$, which are negligible unless $\nu_j \gtrsim 1/\lambda_{\min}$. On a quasi-uniform mesh the eigenvalues ν_j of Ω grow like $\delta\,\omega_j^4$ with $\omega_j \asymp j$ the frequency of u_j (Green and Silverman, 1994), so $\nu_j \gtrsim 1/\lambda_{\min}$ corresponds to $\omega_j \gtrsim 1/\beta_{\min} = (\log n)/h$. The coordinates of a_x in that part of the basis are

the inner products of the kernel $L_h''(x - \cdot)$, sampled on the mesh, with the high-frequency eigenvectors, and are bounded by the Fourier transform of L_h'' at frequency ω_j, which is $\omega_j^2(\frac{8}{3}e^{-h^2\omega_j^2/2} - 2e^{-h^2\omega_j^2} + \frac{1}{3}e^{-2h^2\omega_j^2}) \le Ch^{-2}e^{-(\log n)^2/2}$ there, up to the interpolation and discretization errors already accounted for in (A.2). Hence $\sup_{\lambda\le\lambda_{\min}} \|((I + \lambda\Omega)^{-1} - I)a_x\|^2 \le Cn\,h^{-4}e^{-(\log n)^2}$, which is $o(n^{-2})$, and the sub-Gaussian maximal inequality over the net of Step 2 shows that $\sup_{x,\ \lambda\le\lambda_{\min}} |Z_\lambda(x) - Z_0(x)| = o_p(1)$. Combining the two ranges of λ with (A.3),

$$\sup\{|Z_\lambda(x)| : x \in [A', B'],\ 0 \le \lambda \le \lambda_m\} \ \to_p\ 0.$$

Step 6: the widened bandwidth. The rule (11) sets $h = h_0(1 + \rho_0)^{1/6}$ with $\rho_0 = \rho(h_0)$. Step 3 at the deterministic bandwidth h_0 gives $\rho_0 = O_p(1)$, so for every $\epsilon > 0$ there is a constant A_ϵ with $\mathbb{P}(h_0 \le h \le A_\epsilon h_0) \ge 1 - \epsilon$ for all large m. Every bound in Steps 1 to 5 holds uniformly over $h \in [h_0, A_\epsilon h_0]$: the deterministic bounds because their constants depend on h only through $h \to 0$ and $\bar\delta/h \le \bar\delta/h_0 \to 0$, and the noise bounds by one more chaining coordinate. For the latter, $\partial_{\log h} L_h''(u) = -(3L_h''(u) + uL_h'''(u))$ is a kernel of the same type as L_h'' and satisfies (K2) with the same constants, so the weights $w_i(x)$ are Lipschitz in $\log h$ with proxy $C\tau^2/(nh^5)$ per unit of $\log h$; covering $[\log h_0, \log h_0 + \log A_\epsilon]$ by $O(1)$ cells adds a bounded factor to the net sizes in Steps 2 and 5 and leaves the rates unchanged. The penalty at the widened bandwidth is (12) with $\rho = \rho(h)$, and Step 3 applied at each h in the range, uniformly by the same chaining, gives the event E_m with $\mathbb{P}(E_m^c) \le \epsilon + o(1)$.

Conclusion. On E_m the random penalty lies in $[0, \lambda_m]$ and the random bandwidth in $[h_0, A_\epsilon h_0]$, so Step 4 bounds the deterministic part and Step 5 the noise part of (A.1) at the random penalty and bandwidth; $\mathbb{P}(E_m^c) \le \epsilon + o(1)$ by Steps 3 and 6, and ϵ was arbitrary. Multiplying by $-F$ gives $\|\hat f_{\mathbb{Q}}^{\text{th}} - f_{\mathbb{Q}}\| \to_p 0$ on $[A', B']$, which was an arbitrary compact subset of (A, B). □

Remarks on the proof. The sub-Gaussian assumption is used only in the maximal inequalities; with second moments alone, Chebyshev's inequality over the net of Step 2 would require $nh^7 \to \infty$, which the rule (10) does not deliver at the rate $n^{-1/9}$. The exponential localization in (S2) is what keeps the variance at order $1/(nh^5)$ without a factor involving the mesh ratio, and quasi-uniformity enters only there, in (S4), and in the eigenvalue growth used in Step 5. Nothing in the argument uses the specific form of the thriced kernel beyond (K1)–(K2), nor the number of passes of the spline beyond p being a polynomial that maps $[0, 1]$ into itself with $p(0) = 0$ and a bounded derivative; the same proof gives consistency for the twiced and plain Gaussian kernels and for the plain and twiced splines.

Finally, Step 3 shows why the rule (12) does not disturb consistency: the penalty is h^4 times a quantity that is $O_p(1)$ plus a noise term of order $(nh^7)^{-1/2}$, and the resulting half-width β is below h up to a logarithm whether or not nh^7 diverges; Step 6 shows the same for the widening, which multiplies h_0 by a factor that is $O_p(1)$.

Appendix B. Computation

Equations (4) and (6) ask for one Gaussian convolution of the slope of a cubic spline at three bandwidths, read at every strike. Three evaluations are implemented, and on the designs of Sections 5 and 6 they agree past the printed precision of every table. The first sums the closed-form cell integrals: on each knot interval the slope is a quadratic, the integrals of 1, u and u^2 against ϕ and Φ are elementary, and the cost grows with the number of knots times the number of strikes. The second samples the spline, held flat outside the knots, on a grid one sixty-fourth of the smaller of the median knot spacing and h, padded by ten widths of the widest kernel, and takes one real discrete Fourier transform; (5) is then a single multiplier, differentiation is multiplication by is, and the level is the inverse transform minus the value at the left knot. The third keeps the transform but replaces the fine samples by the exact moments through degree eight of the spline and of its second derivative over boxes no wider than one bandwidth, which are closed form because both integrands are polynomials; the transform of the piecewise polynomial is the sum of nine moment transforms, the value and the derivative are packed into one complex inverse transform, and the jumps at the ends of the strip are added as the kernel itself. Table B.13 times the three on a late-2024 MacBook Pro with an Apple M4 chip, with the penalty fixed at zero, one warm-up call discarded, and the minimum of five runs recorded. The full rule reads the roughness twice, at h_0 and at h, and fits the spline three times; with the box-moment transform it takes 3 to 8 milliseconds on these chains, at most 3.4 times the penalty-zero figure.

Table B.13: Wall-clock seconds for one evaluation of the density and the calls at the quoted strikes, penalty zero. Naive sums the cell integrals, FFT transforms the sampled spline, FFT+ME transforms the box moments. Ratios divide the raw times before rounding.

				Speedup vs. Naive	
Design	Naive	FFT	FFT+ME	FFT	FFT+ME
Heston, dense	0.125	0.004	0.004	30.1	35.7
Heston, sparse	0.013	0.003	0.003	4.7	5.0
Variance gamma, dense	0.123	0.004	0.005	27.9	22.7
Variance gamma, sparse	0.013	0.003	0.003	4.5	4.5
SPX 18 Dec 2026	0.269	0.005	0.003	52.8	103.9
SPX 19 Mar 2027	0.100	0.004	0.002	26.4	42.4
NDX 18 Dec 2026	0.161	0.003	0.002	47.9	69.8
RUT 18 Dec 2026	0.036	0.003	0.002	11.7	17.9

Appendix C. The rules of the source papers on the listed chains

Table C.14 repeats Table 11 for the bandwidth rules that the source papers propose. The Appendix A bandwidth of Aït-Sahalia and Lo, derived for a pooled sample of 14,431 options and carried to a single maturity, is about one standard deviation of moneyness and flattens the smile. The plug-in of Aït-Sahalia and Duarte uses the level rate $n^{-1/5}$ with a global quartic pilot. The leave-one-out width of Grith et al. interpolates the call in sample but prices the held-out wing strikes badly, because a local cubic in call space through four or five deep-in-the-money quotes of level several thousand cannot resolve a premium of half a point; their doubled width is what their figures use.

Appendix D. Further listed chains

Table D.15: Nested even/odd hold-out error in half-spread units on every further standard monthly chain, cross-validated methods. The row minimum is in bold.

Chain	n	T	Ours, rule	Ours, CV	PCA	AS–Lo	AS–D	GHS IV	GHS call
SPX 2026-10-16	438	0.06	**0.08**	0.08	0.54	0.11	0.27	0.09	0.24
SPX 2026-11-20	445	0.16	0.09	**0.09**	0.39	0.17	0.31	0.11	0.79
SPX 2027-01-15	395	0.31	0.08	**0.08**	0.27	0.27	0.31	0.15	1.73
SPX 2027-02-19	338	0.41	0.10	**0.09**	0.32	0.34	0.65	0.24	2.09
SPX 2027-04-16	217	0.56	0.34	0.34	0.49	0.43	1.18	**0.29**	2.75

Chain	n	T	Ours, rule	Ours, CV	PCA	AS–Lo	AS–D	GHS IV	GHS call
SPX 2027-05-21	214	0.66	0.10	**0.09**	0.30	0.42	0.82	0.39	3.28
SPX 2027-06-17	235	0.73	0.16	**0.15**	0.38	0.44	0.96	0.34	1.70
SPX 2027-07-16	209	0.81	0.21	**0.13**	0.40	0.76	1.17	0.84	5.54
SPX 2027-08-20	207	0.91	0.33	**0.14**	0.54	0.77	1.11	1.03	3.87
SPX 2027-09-17	207	0.98	0.54	**0.45**	0.59	0.63	0.95	0.46	0.96
SPX 2027-10-15	198	1.06	0.13	**0.11**	1.56	0.31	0.40	0.22	0.46
SPX 2027-12-17	143	1.23	0.43	0.40	0.38	0.45	0.86	**0.28**	0.59
SPX 2028-01-21	198	1.33	0.23	**0.15**	1.66	0.33	0.29	0.23	0.28
SPX 2028-06-16	199	1.73	0.34	**0.17**	1.76	0.33	0.27	0.24	0.32
NDX 2026-10-16	269	0.06	0.08	**0.07**	0.12	0.12	0.25	0.12	0.79
NDX 2026-11-20	252	0.16	0.10	**0.09**	0.16	0.19	0.16	0.13	0.78
NDX 2027-01-15	104	0.31	0.06	**0.06**	0.11	0.39	0.62	0.09	0.97
NDX 2027-02-19	95	0.41	0.09	**0.09**	0.13	0.36	0.61	0.10	0.99
NDX 2027-03-19	110	0.48	**0.07**	0.07	0.13	0.34	0.38	0.10	1.03
NDX 2027-04-16	83	0.56	**0.07**	0.10	0.17	0.46	0.99	0.10	1.00
NDX 2027-05-21	68	0.66	0.35	0.53	0.59	0.72	1.43	**0.08**	0.48
NDX 2027-06-17	117	0.73	0.15	0.23	0.37	0.26	0.34	**0.08**	0.27
NDX 2027-09-17	96	0.98	0.14	0.16	0.55	0.21	0.32	**0.07**	0.24
NDX 2027-12-17	117	1.23	0.17	0.17	0.22	0.15	0.15	**0.11**	0.15
RUT 2026-10-16	219	0.06	0.08	**0.08**	0.26	0.13	68.62	0.16	19.22
RUT 2026-11-20	203	0.16	**0.09**	0.09	0.30	0.24	0.44	0.17	4.13
RUT 2027-03-19	131	0.48	0.16	0.15	0.38	0.29	0.56	**0.10**	0.43
RUT 2027-06-17	69	0.73	**0.03**	0.06	0.19	0.44	2.60	0.04	0.19
DJX 2027-03-19	44	0.50	0.09	0.12	0.24	0.24	0.58	**0.04**	0.19
DJX 2027-06-17	57	0.74	0.05	0.06	0.16	0.22	0.54	**0.05**	0.11
DJX 2027-09-17	41	0.99	0.32	0.16	0.24	0.15	0.26	**0.03**	0.06

Code and Data Availability

A Python implementation of the estimator, the comparison methods, and the scripts that produce every table and figure are available at `https://github.com/s-broda/RND`. The four working slices of Section 6.1 are included; the further chains of Appendix D are not, under Cboe's data policies.

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Table C.14: Listed chains, bandwidth rules of the source papers. Columns as in Table 11.

		Hold-out			In-sample				
Chain	Method	pts	hs	bp	hs	Mass	Neg.	Modes	TV
SPX Dec 2026	AS–Lo, App. A	37.922	58.79	711	48.82	1.00	-0.000	1	1.00
	AS–Duarte, plug-in	5.385	7.66	43	6.09	1.00	-0.000	2	1.01
	GHS IV, LOO	0.074	0.42	24	0.06	1.03	0.000	3	1.32
	GHS IV, 2×LOO	**0.055**	**0.29**	**19**	0.06	1.01	0.000	1	1.12
	GHS call, LOO	14.653	111.24	1217	0.06	0.98	0.000	23	3.36
	GHS call, 2×LOO	9.964	79.39	874	0.06	0.99	0.000	2	1.12
SPX Mar 2027	AS–Lo, App. A	89.227	120.63	969	100.15	1.00	-0.000	1	1.00
	AS–Duarte, plug-in	9.965	12.09	57	9.54	1.00	-0.000	1	1.00
	GHS IV, LOO	**0.091**	0.48	41	0.07	1.00	-0.000	1	1.00
	GHS IV, 2×LOO	0.282	**0.36**	**19**	0.20	1.00	0.000	1	1.00
	GHS call, LOO	18.635	136.69	1101	0.06	0.99	-0.000	1	1.04
	GHS call, 2×LOO	1.339	1.80	480	0.19	1.00	-0.000	1	1.00
NDX Dec 2026	AS–Lo, App. A	241.766	15.20	982	11.23	1.00	-0.000	1	1.00
	AS–Duarte, plug-in	19.189	1.20	**40**	0.99	1.00	0.000	4	1.11
	GHS IV, LOO	1.248	0.15	67	0.08	1.03	0.000	14	3.02
	GHS IV, 2×LOO	**1.236**	**0.13**	57	0.09	1.03	0.000	6	1.17
	GHS call, LOO	79.411	17.11	1081	0.06	0.88	0.000	13	2.81
	GHS call, 2×LOO	40.791	8.81	884	0.06	0.97	0.000	5	1.15
RUT Dec 2026	AS–Lo, App. A	15.912	28.43	936	22.98	1.00	-0.000	1	1.00
	AS–Duarte, plug-in	2.834	4.79	61	3.76	1.00	0.000	4	1.08
	GHS IV, LOO	**0.039**	**0.10**	14	0.05	1.01	0.000	1	1.00
	GHS IV, 2×LOO	0.116	0.20	**5**	0.08	1.25	-0.000	1	1.00
	GHS call, LOO	7.026	35.13	1266	0.04	1.00	0.000	1	1.05
	GHS call, 2×LOO	0.432	0.82	20	0.10	1.00	0.000	1	1.00

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